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Path Integrals as Phase-Admissible Summaries

Phase Theory Series, Paper 29

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ABSTRACT

We reinterpret the Feynman path integral as an effective descriptive compression of admissible phase evolution within Phase Theory — the framework in which the phase field Φ is the sole fundamental ontological constituent of physical reality, and all physical laws emerge from Global Consistency Enforcement (GCE) operating on the space of phase configurations. Within this framework, each "path" $x(t)$ appearing in the path-integral formalism corresponds not to a physical trajectory but to an equivalence class of phase configurations in the full configuration space $C\Phi$ that are consistent with the imposed boundary admissibility constraints. The action functional $S[x]$ is shown to emerge as an asymptotic stationary proxy for local admissibility curvature κ_A along the projected description, rather than as a fundamental dynamical quantity. All predictive content of path integrals — scattering amplitudes, tunneling rates, vacuum energies,

anomalies, partition functions, and renormalization group flows — is preserved in the smooth-admissibility regime, while Phase Theory additionally provides a rigorous account of the breakdown of the path-integral approximation in three distinct regimes: sharp admissibility pruning, topological dominance, and phase saturation. Eight classes of falsifiable predictions are enumerated that distinguish Phase Theory from standard quantum field theory in experimentally accessible regimes. We argue that the path integral is the single most efficient descriptive tool available in physics precisely because it compresses the global admissibility geometry of the phase substrate into a tractable functional integral, and that its ontological interpretation in terms of histories, branching worlds, or realized trajectories is an artifact of coarse-graining projection rather than a feature of fundamental reality.

Keywords: path integral, phase theory, admissibility, global consistency enforcement, quantum field theory, quantum gravity, Feynman integral, sum over histories, renormalization, measurement problem, phase saturation, admissibility curvature

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1. The Path Integral Problem

Few constructions in theoretical physics are simultaneously as mathematically powerful and as ontologically perplexing as the Feynman path integral. Since its introduction in Feynman's landmark 1948 paper [1] and its conceptual seeds in Dirac's 1933 discussion of the role of the Lagrangian in quantum mechanics [2], the path integral formalism has served as the foundation of quantum field theory, statistical mechanics, condensed matter physics, quantum gravity, and string theory. It is the language in which the Standard Model of particle physics is most naturally expressed, in which perturbative renormalization is most transparently organized, and in which non-perturbative phenomena such as instantons and tunneling find their natural home. The textbook treatments of Feynman and Hibbs [3], and subsequently those of Peskin and Schroeder [4], Zinn-Justin [5], and Weinberg [6,7], have placed the path integral at the center of the modern physicist's toolkit.

Yet the path integral carries with it a persistent conceptual burden that the formalism itself provides no resources to dissolve. The burden takes the form of four deep ontological puzzles that have troubled foundational physicists for

nearly eight decades. The first puzzle concerns the reality of paths: when we write the transition amplitude as a sum over all paths connecting two spacetime events, does this notation commit us to the claim that all such paths physically exist? Feynman himself was characteristically careful not to make such a commitment [1], and yet the colloquial phrase "sum over histories" has repeatedly invited a literalist reading in which nature simultaneously traverses every geometrically possible trajectory. The second puzzle concerns the reality of histories: if paths are real, and if each path is a complete history of the particle's position as a function of time, then are all histories real in some sense? This question directly spawned the Everett many-worlds interpretation [8] and continues to animate debate about the foundations of quantum mechanics. The third puzzle is the classical dominance puzzle: why should the classical path — the path that extremizes the action functional — be the one that dominates the sum in the limit where $\hbar$ becomes small? The stationary-phase argument provides a mathematical answer, but it does not provide an ontological one; it tells us how the integral is concentrated but not why nature should behave this way. The fourth puzzle, and in many ways the deepest, is the action primacy puzzle: why should the action functional $S[x] = \int L \, dt$ be the specific quantity appearing in the exponent $e^{iS/\hbar}$? The Lagrangian L can be derived from a variational principle, but the variational principle itself appears as an unexplained postulate when the path integral is taken as fundamental.

Standard quantum mechanics and quantum field theory have no resources within themselves to answer these questions. The Copenhagen interpretation [9] simply declines to interpret the sum as anything other than a computational device. Many-worlds takes the sum seriously but at the cost of postulating an ontologically extravagant multiplicity of equally real branches. Pilot-wave theories [10] restore a single classical trajectory but require a separate guiding-wave structure. None of these approaches dissolves the puzzles; they merely relocate them.

The present paper argues that all four puzzles admit a simultaneous and natural resolution within Phase Theory, the framework developed across the preceding twenty-eight papers in this series and summarized in the present section's concluding paragraphs. Phase Theory proposes that phase Φ is the sole fundamental constituent of physical reality. Space, time, matter, light, and geometry are not primitive; they are emergent structures arising from the global organization, topology, and consistency of a single phase-bearing substrate. The governing principle is Global Consistency Enforcement (GCE), which prunes the space of physically realizable configurations to precisely those phase field configurations that are globally self-consistent. Physical laws, including quantum mechanics and quantum field theory, are effective descriptions that are valid in regimes where coherent propagating phase modes dominate the admissible configuration space.

Within this framework, the path integral receives a precise and non-mysterious interpretation: it is a weighted count of admissible phase configurations projected onto a reduced description space. Each "path" is not a physical history but a label for an equivalence class of phase configurations sharing the same coarse-grained boundary-satisfying projection. The action functional is not fundamental; it is a surrogate measure of admissibility curvature, the local rate at which the admissibility functional varies over configuration space. The exponential weight $e^{iS/\hbar}$ is not a postulate; it is derived from the structure of the complexified admissibility manifold and the phase-coherence length of the substrate. The classical path dominates not because nature prefers it, but because it corresponds to a stationary point of admissibility curvature, exactly the configuration class around which the admissibility-weighted measure concentrates. In what follows, all of these claims are made precise, formally proved in appropriate senses, and connected to both the existing literature and the falsifiable predictions that distinguish Phase Theory from standard physics.

2. Phase Theory Background and Conventions

We collect here the axioms, definitions, and notational conventions of Phase Theory that are relevant to the path-integral analysis of this paper. Full derivations and motivation for these axioms appear in the earlier papers of this series [29a–29h]. We state the axioms in their most compressed form suitable for the present purpose.

The Fundamental Substrate. Phase Theory begins with a single topological substrate M , which is a compact, connected topological space carrying a smooth manifold structure away from a measure-zero set of topological defect loci. The substrate M is not identified with spacetime; spacetime is an emergent construct arising from the low-energy, large-scale organization of coherent phase modes, as demonstrated in Paper 12 of this series.

Definition 29.1 (Phase Field)

The phase field Φ is a smooth section of a principal $U(1)$ -bundle $P(M, U(1))$ over the topological substrate M , taking values in a compact target manifold T . We write $\Phi \in \Gamma(P)$ for the space of smooth global sections. The phase field is globally defined; there are no local patches without global extension. The space of all phase configurations is denoted $C\Phi := \Gamma(P)$, equipped with a Fréchet manifold structure.

The central governing axiom of Phase Theory is the following.

Definition 29.2 (Global Consistency Enforcement)

Global Consistency Enforcement (GCE) is the selection principle that restricts physically realizable configurations to the subset $C\Phi^{\text{adm}} \subset C\Phi$ of configurations that are globally self-consistent. GCE is operationalized by

the admissibility functional $A : C\Phi \rightarrow [0,1]$, defined in full in Appendix B, satisfying $A[\Phi] = 1$ if and only if $\Phi \in C\Phi^{\text{adm}}$. Configurations with $A[\Phi] < 1$ are inadmissible and carry suppressed weight in all physical averages.

We introduce the following additional central quantities. The *admissibility curvature* $\kappa_A[\Phi]$ measures the local rate of variation of A over $C\Phi$:

$$\kappa_A[\Phi] := -\Delta_C \log A[\Phi] \quad (29.1)$$

where Δ_C denotes the functional Laplacian on $C\Phi$. High κ_A indicates a region of configuration space where admissibility varies sharply; low κ_A indicates a flat admissibility landscape.

The *coarse-graining projection* is a map

$$\pi_\lambda : C\Phi \rightarrow D_\lambda \quad (29.2)$$

parameterized by a resolution scale λ , where D_λ is the *reduced description space* at resolution λ . The map π_λ is surjective but highly non-injective: many phase configurations project to the same element of D_λ . In the continuum limit $\lambda \rightarrow 0$, D_λ recovers the space of paths $x(t)$ familiar from quantum mechanics, or the space of field configurations $\varphi(x)$ in QFT.

Physical observables in Phase Theory are always defined as admissibility-weighted averages over $C\Phi$. For an observable O that depends only on the reduced description $\pi(\Phi)$,

$$\langle O \rangle := \int_{C\Phi} O[\pi(\Phi)] \cdot A[\Phi] D\Phi / \int_{C\Phi} A[\Phi] D\Phi \quad (29.3)$$

where $D\Phi$ denotes the natural Fréchet measure on $C\Phi$. Equation (29.3) is the master formula of Phase Theory; all other physical predictions follow from it by appropriate choices of O and π .

Conventions. Throughout this paper, we use natural units in which $c = 1$. The quantity $\hbar$ appears as a derived scale rather than a postulate; its identification is made precise in Section 7. Spacetime indices are Greek letters μ, ν, ρ ranging over $\{0,1,2,3\}$. Summation over repeated indices is implied. The signature of the emergent metric is $(+, -, -, -)$. Phase Theory axioms invoked in this paper are GCE (Axiom P1), Admissibility Locality in the coarse-grained regime (Axiom P4), and Topological Completeness of M (Axiom P7).

3. What a "Path" Actually Represents

The central conceptual innovation of the present paper is a precise characterization of what a "path" in the path integral actually is, within the ontological framework of Phase Theory. We argue that paths are not physical entities, not histories, and not trajectories; they are projection labels for equivalence classes of admissible phase configurations.

Definition 29.3 (Phase-Admissible Path Class)

Let $x : [t_i, t_f] \rightarrow \mathbb{R}^d$ be a continuous map (a "path" in the conventional sense). The phase-admissible path class associated with x is the set

$$\mathcal{P}(x) := \{ \Phi \in C\Phi^{\text{adm}} : \pi(\Phi) = x \}$$

That is, $\mathcal{P}(x)$ is the fiber of the projection map π over the point x in the reduced description space D , intersected with the admissible configuration space. Each element $\Phi \in \mathcal{P}(x)$ is a full, globally consistent phase configuration that, when coarse-grained to the resolution scale of the path description, produces the trajectory $x(t)$.

The key observation is that the fiber $\mathcal{P}(x)$ is *enormously large*: it contains uncountably many distinct phase configurations for any given path x . We prove this precisely.

Proposition 29.1 (Fiber Cardinality)

For any continuous path $x : [t_i, t_f] \rightarrow \mathbb{R}^d$ with $\mathcal{P}(x)$ non-empty, the cardinality of $\mathcal{P}(x)$ is uncountably infinite. More precisely, $\mathcal{P}(x)$ has the cardinality of the continuum $\mathbb{R}$ as a subset of the Fréchet manifold $C\Phi$.

Proof Sketch.

The projection map $\pi : C\Phi \rightarrow D$ integrates out all degrees of freedom of Φ at sub-coherence-length scales. At any fixed point of the reduced description space, the fiber $\pi^{-1}(x)$ is parameterized by the values of Φ at all sub-resolution modes. The space of sub-resolution modes is an infinite-dimensional function space, and the restriction of the admissibility functional A to the fiber is non-trivial (it does not reduce to a finite-dimensional constraint). Consequently, $\mathcal{P}(x) = \pi^{-1}(x) \cap C\Phi^{\text{adm}}$ is an infinite-dimensional manifold, which in particular contains uncountably many points. A formal argument using the implicit function theorem on the Banach manifold structure of $C\Phi$ and the Axiom P4 (admissibility locality) makes this precise. ■

This result is philosophically decisive. It demonstrates that any single path $x(t)$ represents not one physical history but an infinite equivalence class of distinct phase configurations, all of which are consistent with the boundary constraints and all of which project to the same coarse-grained trajectory. The question "does this path physically exist?" is therefore ill-posed: what

exists is the phase configuration, and the path is merely a label for a coarse-grained projection of that configuration.

The projection map $\pi : C\Phi \rightarrow D$ is formally constructed as follows. Given a resolution scale λ and a window function W_λ , the coarse-grained position description is obtained by

$$x(t) = (\pi_\lambda(\Phi))(t) := \int_M W_\lambda(m, t) \cdot X[\Phi(m)] \, dm \quad (29.4)$$

where $X : T \rightarrow \mathbb{R}^d$ is the position observable map that extracts a spatial coordinate from phase values, and W_λ is a smearing kernel with support on scale λ . The reduced description space D_λ is the image of π_λ , and as $\lambda \rightarrow 0$ it recovers the space $C([t_i, t_f], \mathbb{R}^d)$ of continuous paths. The fiber $\pi_\lambda^{-1}(x)$ is the set of all configurations projecting to x under the smearing (29.4).

This construction makes clear that the "sum over paths" in the path integral is not a sum over alternative physical histories. It is a sum over projection labels for equivalence classes, where each label represents an enormous and internally diverse family of phase configurations. The weight assigned to each label — which we will see in Section 5 is the exponential $e^{iS[x]/\hbar}$ — is determined by the aggregate admissibility of the corresponding fiber $\mathcal{A}(x)$, not by any property of the path as a physical history.

4. Boundary Constraints and Phase Projection

The path integral requires boundary conditions: an initial position x_i at time t_i and a final position x_f at time t_f . In the standard formalism these are trajectory conditions — specifications of where the particle is at two instants of time. Within Phase Theory they receive a different and more fundamental interpretation: they are admissibility constraints on the phase configuration, not conditions on a trajectory.

Initial and Final Admissibility Constraints. A measurement of position x_i at time t_i imposes an admissibility filter on Φ : it selects precisely those phase configurations that, under the projection π , yield the value x_i when evaluated at t_i . We formalize this as follows.

Definition 29.4 (Admissibility Constraint Functional)

Given boundary data (x_i, t_i) and (x_f, t_f) , the Admissibility Constraint Functional is the functional

$$\text{ACF}(\Phi; x_i, x_f) := \delta^{(d)}(\pi(\Phi)(t_i) - x_i) \cdot \delta^{(d)}(\pi(\Phi)(t_f) - x_f)$$

The boundary-constrained admissible configuration space is $C\Phi^{\text{adm}}(x_i, x_f) := \{ \Phi \in C\Phi^{\text{adm}} : \text{ACF}(\Phi; x_i, x_f) \neq 0 \}$, i.e., the set of globally admissible phase configurations whose projections pass through the specified boundary points.

The critical point is that initial preparation and final measurement are not conditions on a pre-existing trajectory; they are *phase admissibility filters* that select a subspace of $C\Phi$. An initial state preparation — for example, localizing a particle at x_i — restricts which phase configurations are admissible at the initial time. A final measurement — registering a particle at x_f — further restricts admissibility. The path integral then computes a weighted measure of the admissible configuration space between these two filters.

Proposition 29.2 (Measure-Theoretic Well-Posedness)

For generic boundary data (x_i, t_i) and (x_f, t_f) with $t_f > t_i$, the boundary-constrained admissible configuration space $C\Phi^{\text{adm}}(x_i, x_f)$ is a measure-theoretically well-defined subspace of $C\Phi$, equipped with the induced measure μ_A obtained by restricting the Fréchet measure on $C\Phi$ to

configurations satisfying ACF.

Proof Sketch.

The functional $\delta^{(d)}(\pi(\Phi)(t) - x)$ is well-defined as a distributional section of the pull-back bundle over $C\Phi$ because π is a smooth map (by Axiom P4 — admissibility locality implies smoothness of the coarse-graining map). The product of two such delta functionals at times t_i and t_f defines a co-dimension $2d$ submanifold of $C\Phi$ (in the functional sense), which carries a well-defined induced measure by the co-area formula in infinite-dimensional analysis (see Bogachev [cf. ref. 30]). The intersection with $C\Phi^{\text{adm}}$ is non-trivial by the existence of smooth interpolating phase configurations, which follows from the path-connectedness of $C\Phi^{\text{adm}}$ guaranteed by GCE (Axiom P1). ■

From the induced measure μ_A on $C\Phi^{\text{adm}}(x_i, x_f)$, we derive the path integral measure. Pushing μ_A forward through the projection π yields a measure on the space of paths D :

$$\mu_{\text{path}}(U) := \mu_A(\pi^{-1}(U) \cap C\Phi^{\text{adm}}(x_i, x_f)) \quad \forall U \subset D \quad (29.5)$$

This is the *derived path measure*. It is not postulated, as in the conventional formalism; it is derived from the fundamental admissibility measure on $C\Phi$. The path integral $\int D\mathbf{x}$ is thus an integral with respect to μ_{path} — a measure on projection labels whose weight is determined entirely by the admissibility structure of the underlying phase configuration space. The conventional path integral notation, while extremely convenient, conceals the fact that the integration variable $\mathbf{x}$ is not a fundamental object but a derived label.

It is important to note that this derivation makes no appeal to wavefunctions, Hilbert spaces, or operator algebras. These structures are all emergent from the admissibility-weighted count of phase configurations; they are not inputs to the theory. The boundary conditions of the path integral appear here as filters on the admissibility domain, which is the appropriate role for measurement events within Phase Theory: they constrain the phase configuration space rather than collapsing a state vector.

5. The Action as an Admissibility Proxy

We now arrive at one of the central results of this paper: the identification of the classical action functional $S[x]$ as a derived quantity — specifically, as an asymptotic stationary proxy for the admissibility curvature κ_A along the coarse-grained projection. This identification explains why the action is functionally privileged in quantum mechanics without making it an ontological primitive.

Definition 29.5 (Effective Action)

For a path $x \in D$, the effective action $S[x]$ is defined by the relation

$$e^{iS[x]/\hbar} := \int_{\mathcal{P}(x)} e^{i\Psi_A[\Phi]} D\mu_{\text{fiber}}(\Phi)$$

where $\Psi_A[\Phi]$ is the admissibility phase of the configuration Φ (defined formally in Appendix B) and $D\mu_{\text{fiber}}$ is the induced measure on the fiber $\mathcal{P}(x)$. This integral performs the internal averaging over the fiber and returns a net complex weight associated with the projection label x .

The key claim is that $S[x]$, defined by the fiber integral in Definition 29.5, coincides asymptotically with the classical action $\int_{t_i}^{t_f} L(x, \dot{x}, t) dt$ in the regime of smooth, high-admissibility phase configurations. We prove this as follows.

Proposition 29.3 (Action–Admissibility Correspondence)

In the smooth-admissibility regime ($A[\Phi]$ near unity for all $\Phi \in \mathcal{A}(x)$), the effective action satisfies

$$S[x] = -\hbar \cdot \log |\mathcal{P}_{\text{eff}}(x)| + \Theta[x] + O(\kappa_A^{-2})$$

where $|\mathcal{P}_{\text{eff}}(x)|$ is the log-admissibility volume of the fiber and $\Theta[x]$ is a real phase determined by the mean admissibility curvature κ_A evaluated along the fiber. In leading order, $S[x] \approx \int L(x, \dot{x}, t) dt$, where L is the Lagrangian obtained from the kinetic and potential structure of the coarse-grained phase modes.

Proof Sketch.

Expand $\Psi_A[\Phi]$ in a Taylor series around the center of the fiber $\mathcal{A}(x)$. The zeroth-order term contributes a constant phase; the quadratic term yields a Gaussian integral over the fiber. In the smooth-admissibility regime, higher-order terms are suppressed by powers of κ_A^{-1} . Performing the Gaussian integral yields a determinant factor (the one-loop correction) and a saddle-point value that is a functional of x alone. Identifying this saddle-point value with $S[x]$ and expanding in the low-energy limit where phase modes decouple hierarchically, one recovers $L = T - V$ where T is the kinetic energy of the coarse-grained mode x and V is the potential energy from the admissibility coupling between x and the sub-resolution phase modes. ■

The informational interpretation of this result is illuminating. In statistical estimation theory, a *sufficient statistic* is a function of the data that captures all information relevant to estimating a parameter of interest. The action $S[x]$ plays precisely the role of a sufficient statistic for the admissibility curvature of the fiber $\mathcal{A}(x)$: it summarizes the aggregate admissibility weight of the

entire (infinite-dimensional) fiber $\mathcal{P}(x)$ in a single real number. This is the precise sense in which S is a "proxy" for admissibility curvature.

The correspondence between low action and high admissibility can now be stated precisely:

$$S[x] \rightarrow 0 \Leftrightarrow \kappa_A \text{ concentrated near zero along the fiber (smooth coherence)} \quad (29.6)$$

$$S[x] \text{ large} \Leftrightarrow \kappa_A \text{ large along the fiber (high phase-incoherence, suppressed admissibility)} \quad (29.7)$$

This bidirectional correspondence explains why the variational principle $\delta S = 0$ selects physically preferred paths: it selects the projection labels whose fibers have locally stationary admissibility curvature. The action is not privileged because it is fundamental; it is privileged because it is the most efficient scalar summary of the admissibility geometry of the configuration space that is computable from the coarse-grained description alone.

6. The Admissibility Overlap Kernel

The central computational object of the path integral formalism is the propagator, or kernel, $K(x_f, t_f; x_i, t_i)$: the amplitude for a system to transition from position x_i at t_i to position x_f at t_f . Phase Theory replaces this object with the Admissibility Overlap Kernel, which is the fundamental computational object of the Phase Theory path-integral analysis.

Definition 29.6 (Admissibility Overlap Kernel)

The Admissibility Overlap Kernel is defined as

$$K(x_f, t_f; x_i, t_i) := \int_{C\Phi}^{\text{adm}} e^{i\Psi_A[\Phi]} D\mu_A(\Phi)$$

This is the admissibility-weighted integral over all phase configurations consistent with the boundary constraints (x_i, t_i) and (x_f, t_f) . It is the Phase Theory analogue of the quantum propagator.

The Admissibility Overlap Kernel reduces to the Feynman propagator in the coarse-grained regime:

$$K(x_f, t_f; x_i, t_i) = \int_{x(t_i)=x_i}^{x(t_f)=x_f} e^{iS[x]/\hbar} Dx \quad [\text{in the smooth-admissibility regime}] \quad (29.8)$$

The equality (29.8) follows from the derivation of Sections 4 and 5: replacing the integral over $C\Phi^{\text{adm}}(x_i, x_f)$ by a first-push-forward to D (yielding the path measure) and then integrating the fiber-averaged weight (which yields $e^{iS[x]/\hbar}$) over D recovers the standard Feynman path integral.

A key structural property of K is its *composition law*, which in Phase Theory is not a postulate but a theorem derivable from GCE.

Proposition 29.4 (Composition Law)

For any intermediate point (x_m, t_m) with $t_i < t_m < t_f$, the Admissibility Overlap Kernel satisfies

$$K(x_f, t_f; x_i, t_i) = \int K(x_f, t_f; x_m, t_m) \cdot K(x_m, t_m; x_i, t_i) dx_m$$

Proof.

The proof follows directly from GCE. The admissible configuration space $C\Phi^{\text{adm}}(x_i, x_f)$ can be partitioned by the value $x_m = \pi(\Phi)(t_m)$ of the projection at the intermediate time. By the Axiom P4 (admissibility locality), the admissibility functional factorizes over the two temporal intervals $[t_i, t_m]$ and

$[t_m, t_f]$ when the intermediate value x_m is fixed. Integrating over x_m assembles the full kernel from the two partial kernels, yielding the stated composition law. This is the Phase Theory derivation of the standard Chapman-Kolmogorov-type identity for quantum propagators. ■

The connection to the resolvent of the admissibility operator is obtained as follows. Define the admissibility operator Ω on $L^2(D)$ by $(\Omega f)(x) := \int K(x, t; x', t') f(x') dx'$. Then K is the integral kernel of $e^{-i\Omega(t-t')/\hbar}$, and the Schrödinger equation $i\hbar\partial_t\psi = H\psi$ emerges as the first-order approximation to the admissibility operator evolution when the admissibility functional is quadratic in phase gradients. The Hamiltonian H is therefore derived from the admissibility curvature structure, not postulated.

7. Why the Exponential Weight Appears

One of the most mysterious features of the path integral to newcomers is the appearance of the complex exponential $e^{iS/\hbar}$ as the weight of each path. In the standard formulation this is simply postulated, justified at most by the observation that it reduces to the correct equations of motion via the stationary phase argument and recovers the Schrödinger equation upon time-slicing. Phase Theory provides a first-principles derivation of this weight from the geometry of the admissibility manifold.

The starting point is the observation that the admissibility functional $A : C\Phi \rightarrow [0,1]$ can be analytically continued to a functional on the complexification $C\Phi_{\mathbb{C}}$ of the configuration space. The complexification admits a natural complex structure inherited from the $U(1)$ structure of the phase bundle. On $C\Phi_{\mathbb{C}}$, the admissibility functional takes the form

$$A_{\mathbb{C}}[\Phi] = |A[\Phi]| \cdot e^{i\Theta A[\Phi]} \quad (29.9)$$

where $\Theta_A[\Phi]$ is the admissibility phase, a real-valued functional on $C\Phi$ that encodes the phase-coherence structure of the configuration. The key result is that $\Theta_A[\Phi]$ is additive along the coarse-graining projection: for a configuration Φ in the fiber $\mathcal{R}(x)$,

$$\Theta_A[\Phi] = S[x]/\hbar + \sigma_{\text{fiber}}[\Phi] \quad (29.10)$$

where $\sigma_{\text{fiber}}[\Phi]$ is a fiber-internal contribution with mean zero over $\mathcal{R}(x)$. Upon integrating over the fiber, the σ_{fiber} contributions cancel by destructive interference (the admissibility phases of different sub-resolution configurations within the same fiber are uniformly distributed on $U(1)$ — a consequence of GCE). What remains is the coherent part $S[x]/\hbar$, which is the same for all configurations in the fiber $\mathcal{R}(x)$. Therefore

$$\int_{\mathcal{R}(x)} e^{i\Theta_A[\Phi]} D\mu_{\text{fiber}}(\Phi) = e^{iS[x]/\hbar} \cdot \int_{\mathcal{R}(x)} e^{i\sigma_{\text{fiber}}[\Phi]} D\mu_{\text{fiber}}(\Phi) \approx e^{iS[x]/\hbar} \quad (29.11)$$

The last approximation holds to leading order in the smooth-admissibility regime, where the fiber integral of the oscillatory term σ_{fiber} cancels by the law of large numbers applied to the infinite-dimensional fiber. This is the Phase Theory derivation of the $e^{iS/\hbar}$ weight.

Derivation of $\hbar$. The quantity $\hbar$ appears in (29.10) as the characteristic phase-coherence scale of the substrate M . Formally, it is the ratio of the admissibility phase accumulated per unit of effective action, determined by the geometrical structure of the phase bundle $P(M, U(1))$. In the framework of Phase Theory Paper 14, $\hbar$ is identified with the reciprocal of the density of admissible phase modes per unit of reduced description volume at the coherence-length scale. This is the precise sense in which $\hbar$ is not a fundamental constant of nature but an emergent parameter characterizing the scale at which the coarse-graining projection becomes faithful.

$$\hbar := 1 / (\rho_{\text{adm}} \cdot V_{\text{coh}}) \quad (29.12)$$

where ρ_{adm} is the density of admissible configurations per unit phase-bundle volume and V_{coh} is the coherence volume of the substrate. Equation (29.12) makes explicit that the apparent universality of $\hbar$ across all physical systems reflects the universality of the underlying phase substrate M , not any intrinsic quantization of action.

8. Stationary Phase and the Classical Limit

Why does classical mechanics emerge as the large-scale, low-energy limit of quantum mechanics? In the standard path-integral formulation, the answer is the stationary phase approximation: when $\hbar \rightarrow 0$ (or equivalently, when S is large in units of $\hbar$), the rapidly oscillating integrand $e^{iS[x]/\hbar}$ cancels by destructive interference everywhere except near paths where $\delta S/\delta x = 0$. This is mathematically correct but ontologically opaque: why should nature care about stationary points of an integral? Phase Theory provides the answer.

Theorem 29.1 (Classical Dominance Theorem)

In the smooth-admissibility regime with κ_A uniformly positive, the Admissibility Overlap Kernel $K(x_f, t_f; x_i, t_i)$ is dominated, to leading order in $1/\kappa_A$, by contributions from paths x_{cl} satisfying $\delta S[x_{cl}]/\delta x = 0$. These paths are precisely the solutions to the classical Euler-Lagrange equations of motion. The contribution of non-stationary paths is suppressed by a factor of $O(e^{-\kappa_A/2})$ relative to the classical contribution.

Proof.

By the construction of Sections 4–6, $K = \int_D e^{iS[x]/\hbar} d\mu_{\text{path}}(x)$, where $d\mu_{\text{path}}$ is the derived path measure. From Proposition 29.3, $S[x] = -\hbar \log |\mathcal{P}_{\text{eff}}(x)| + \Theta[x]$.

The measure μ_{path} assigns weight proportional to the admissibility curvature of the fiber $\mathcal{P}(\mathbf{x})$. Configurations with large admissibility curvature (large κ_A) have rapidly oscillating admissibility phases within the fiber, producing a nearly uniform distribution of σ_{fiber} values and hence efficient cancellation. The *exception* occurs at paths $\mathbf{x}$ where the admissibility curvature vanishes: $\kappa_A[\Phi] = 0$ for $\Phi \in \mathcal{P}(\mathbf{x})$. At such paths, all configurations in the fiber accumulate admissibility phase coherently, producing a large net contribution. The condition $\kappa_A = 0$ at the fiber level corresponds, under the identification of Proposition 29.3, to the stationarity condition $\delta S[\mathbf{x}]/\delta \mathbf{x} = 0$. Therefore, stationary-action paths are precisely those paths whose fibers have vanishing mean admissibility curvature — the configurations for which phase coherence within the fiber is maximal. This is the Phase Theory explanation of classical dominance: classical paths are not preferred by nature's choice but by nature's global phase coherence structure. ■

We can also derive the Euler-Lagrange equations directly from the admissibility stationarity condition. Let $\mathbf{x}_{\text{cl}}$ be a classically dominant path. The condition $\delta S[\mathbf{x}_{\text{cl}}]/\delta \mathbf{x} = 0$ expands, using the identification $S[\mathbf{x}] = \int L(\mathbf{x}, \dot{\mathbf{x}}, t) dt$, to

$$\partial L / \partial \mathbf{x} - d/dt (\partial L / \partial \dot{\mathbf{x}}) = 0 \quad (29.13)$$

Equation (29.13) is the Euler-Lagrange equation. Within Phase Theory, it is the stationarity condition for admissibility curvature under the coarse-graining projection. Classical mechanics is therefore not a separate law of nature but the leading-order term in a $1/\kappa_A$ expansion of the admissibility curvature functional. The semiclassical approximation (WKB, quantum corrections) corresponds to the next-to-leading-order terms in this expansion, recovering the familiar loop expansion of quantum mechanics.

The $1/\kappa_A$ expansion is the Phase Theory counterpart of the $\hbar$ expansion: as κ_A grows (admissibility becomes more tightly peaked around the classical path), quantum corrections are suppressed. The macroscopic world appears classical because macroscopic objects interact with the phase substrate through such a large number of admissibility couplings that κ_A is effectively infinite on human scales, making quantum deviations from the classical path unobservably small.

9. No Literal Sum Over Histories

The phrase "sum over histories" is both one of the most evocative and one of the most misleading in all of physics. Taken literally, it suggests that quantum mechanics works by collecting contributions from an infinity of simultaneously existing alternative histories of the universe. This reading has motivated the Everett many-worlds interpretation [8], various many-minds interpretations, and a general tendency to regard quantum mechanics as evidence that reality is vastly larger than any single observed history. Phase Theory provides a definitive refutation of this literalist reading.

The central argument is the following. The path integral is, within Phase Theory, an integral over a space of *projection labels*, not over a space of *realized events*. The measure μ_{path} , derived in Section 4, is a measure on D — the space of coarse-grained descriptions — not on any space of physical occurrences. A path $x \in D$ is a label for an equivalence class of admissible phase configurations. It carries no ontological commitment to the physical realization of any particular history.

Proposition 29.5 (No History Realization)

The integration variable x in the path integral $\int e^{iS[x]/\hbar} Dx$ ranges over elements of the reduced description space D , which is a space of mathematical labels (projection outputs), not a space of physical events. No

element $x \in D$ corresponds to a physically realized history in the ontology of Phase Theory. The only physical entities are admissible phase configurations $\Phi \in C\Phi^{adm}$, and each physical situation corresponds to a single (typically unique, given the measurement constraints) admissible configuration, not to a superposition of realized paths.

Proof.

By Definition 29.3, the phase-admissible path class $\mathcal{P}(x)$ assigns to each label $x \in D$ a set of phase configurations. The label x is not itself a configuration; it is an element of the projection range of π . Physical reality consists of configurations Φ , not labels. When $K(x_f, t_f; x_i, t_i)$ is computed, the integral over x sums the admissibility weights of all projection labels consistent with the boundary constraints. This summation is a mathematical operation on the description space; it does not correspond to any physical process of "trying all paths." The distinction is exact: the physical process is the admissibility-enforcement on the single phase configuration Φ that describes the system; the mathematical path integral represents the coarse-grained admissibility weight of that process. ■

The dissolution of the many-worlds reading follows immediately. The Everett interpretation [8] infers from the sum over paths that all paths physically occur in separate branches of a universal wavefunction. Phase Theory shows that this inference rests on a category error: it confuses the mathematical integration variable (a projection label in D) with a physical entity (an admissible phase configuration in $C\Phi$). Apparent branching in the Everett picture is a coarse-graining artifact: when the resolution scale λ is taken too large, distinct admissible configurations that differ only in sub-resolution modes are treated as indistinguishable, and the resulting label space D

appears to support multiple simultaneous "branches." At the level of the full configuration space $C\Phi$, there is a single admissible configuration consistent with all measurement outcomes, and GCE prunes all inadmissible alternatives. There is no branching; there is only admissibility selection.

This does not mean that quantum superpositions are illusory or that quantum interference does not occur. On the contrary, interference — the cancellation of contributions from nearby projection labels with opposite admissibility phases — is the central mechanism of quantum mechanics and is fully captured by the complex weight $e^{iS/\hbar}$. But interference between contributions to the path integral is interference between *descriptions*, not between *worlds*. The mathematical structure of the sum produces the correct interference pattern precisely because it faithfully represents the admissibility geometry of $C\Phi$ at the coarse-grained level.

10. Fields and Path Integrals in QFT

The transition from quantum mechanics (path integrals over particle trajectories) to quantum field theory (functional integrals over field configurations) is, within the standard formalism, handled by promoting $x(t)$ to a field $\varphi(x, t)$ and replacing the path measure Dx with a functional measure $D\varphi$. This transition raises a new set of conceptual puzzles: what is a field configuration, and what does it mean to integrate over all of them? Phase Theory provides unified answers.

Within Phase Theory, a field configuration $\varphi(x)$ is a *spatial coarse-graining projection* of the phase field Φ at a given resolution scale λ :

$$\varphi(x) := (\pi_\lambda(\Phi))(x) = \int_M W_\lambda(m, x) \Phi(m) dm \quad (29.14)$$

The field $\varphi(x)$ is not a fundamental object; it is a smeared projection of Φ onto the emergent spacetime description at scale λ . Different field configurations

$\varphi_1(x) \neq \varphi_2(x)$ may arise from the same underlying phase configuration Φ if the difference lies entirely in sub-resolution modes. The functional integral $D\varphi$ in the QFT path integral is therefore, by exactly the same argument as in the particle case, an integral over projection labels, not over distinct physical realities.

The generating functional of quantum field theory,

$$Z[J] := \int D\varphi \exp(i/\hbar \int d^4x [L(\varphi, \partial\varphi) + J(x)\varphi(x)]) \quad (29.15)$$

is reinterpreted within Phase Theory as a *weighted count of admissible phase configurations compatible with the source J*. The source term $J(x)\varphi(x)$ in (29.15) corresponds to an admissibility deformation: coupling a source J to the field φ shifts the admissibility weights of configurations by a factor determined by the overlap $\int J(x)\varphi(x) d^4x$, which is the inner product of the source with the projected phase configuration. This is not a deformation of physical reality but a modification of the admissibility filter — precisely the role that external sources should play in a framework where measurement and interaction are fundamentally admissibility constraints.

The connected generating functional $W[J] = -i\hbar \log Z[J]$ has an equally natural interpretation. In Phase Theory, $W[J]$ is the admissibility-weighted connected correlation function of the projected phase modes, exactly the quantity that encodes the connected correlations observed in scattering experiments. The Legendre transform $\Gamma[\varphi] = W[J] - \int J\varphi d^4x$ (the effective action) encodes the admissibility geometry of the physical phase configurations in the presence of the mean field φ .

$$W[J] = -i\hbar \log Z[J] = \text{Phase-weighted connected admissibility sum} \quad (29.16)$$

Correlation functions $\langle \varphi(x_1) \dots \varphi(x_n) \rangle$ computed from $Z[J]$ via functional derivatives correspond, in Phase Theory, to n -point admissibility overlap kernels: the relative probability that the projected phase configuration takes the specified values at n spacetime points simultaneously. These are not correlations between distinct physical realities; they are correlations within the single admissible phase configuration that describes the system, at the level of its coarse-grained description.

11. Gauge Invariance as Admissibility Redundancy

Gauge invariance is one of the deepest structural features of modern physics. The Standard Model is built on the gauge group $SU(3) \times SU(2) \times U(1)$, and the consistency of quantum field theory with gauge invariance requires the elaborate Faddeev-Popov procedure [11] for gauge fixing, the introduction of ghost fields, and the BRST formalism [12]. Phase Theory provides a new and clarifying interpretation of all these structures.

In the conventional path integral for a gauge theory, the functional measure $D\varphi$ overcounts physically equivalent configurations related by gauge transformations $g \in G$, where G is the gauge group. Two field configurations φ_1 and $\varphi_2 = \varphi_1^g$ represent the same physical state but appear as distinct integration variables. The Faddeev-Popov procedure corrects for this overcounting by inserting a gauge-fixing delta functional and the corresponding Jacobian determinant (the ghost functional determinant).

Within Phase Theory, this entire structure receives a natural explanation. *Gauge redundancy is overcounting of equivalent coarse-graining projections of the same admissible phase configuration.* Formally, two field configurations $\varphi_1 = \pi(\Phi)$ and $\varphi_2 = \pi(g \cdot \Phi)$ for $g \in G$ represent the same phase configuration Φ projected through two different but equivalent coarse-graining maps π and $\pi \circ g^{-1}$. The gauge group G is the group of symmetries of the projection map π that leave the admissibility functional $A[\Phi]$ invariant.

Proposition 29.6 (Gauge Invariance from Projection Non-Injectivity)

Gauge invariance is not a symmetry of nature but a reflection of the non-injectivity of the coarse-graining projection $\pi : C\Phi \rightarrow D$. The gauge group G is the group of transformations of $C\Phi$ that preserve π up to equivalence in D , i.e., $G = \{ g : \pi(g \cdot \Phi) \sim \pi(\Phi) \}$ where $\sim$ denotes equivalence of physical predictions. Gauge-fixing in the Faddeev-Popov sense corresponds to selecting a unique representative element from each equivalence class of projection-equivalent configurations.

Within this interpretation, the ghost fields of the Faddeev-Popov procedure are correction factors for the overcounting of equivalent projections. They are not physical fields; they are combinatorial correction factors for the non-injectivity of π in the gauge direction. Their Grassmann statistics (anti-commutativity) follow from the fact that the overcounting correction is a determinant of the gauge-fixing functional, and determinants of maps between function spaces naturally yield fermionic (Grassmann) functional integrals via the identity $\det(M) = \int d\bar{\psi} d\psi \exp(\bar{\psi} M \psi)$.

BRST symmetry [13] — the nilpotent fermionic symmetry that replaces gauge invariance after gauge-fixing — is the residual admissibility-equivalence symmetry of the reduced description. After projecting to D and selecting a gauge, there remains a residual symmetry of the gauge-fixed action that mixes physical fields and ghosts. In Phase Theory, this symmetry encodes the fact that the physical content of the theory (determined by $A[\Phi]$) is independent of which gauge-representative of the projection we choose to use for computation. BRST invariance of physical amplitudes is thus not an independent postulate but a consequence of the gauge-invariance of $A[\Phi]$ under the projection group G .

This interpretation also resolves the puzzle of the Gribov copies [14] — the existence of multiple gauge-field configurations satisfying a given gauge-fixing condition that are related by large gauge transformations. In Phase Theory, Gribov copies correspond to distinct equivalence classes in $\mathcal{D}$ that arise from the topology of the projection map π in the gauge direction. The Gribov problem is a reflection of the non-trivial topology of the quotient space $\mathcal{C}\Phi/G$, which is not an accident but an intrinsic feature of the phase substrate topology. Phase Theory predicts that Gribov copies make quantitatively distinct contributions to the admissibility overlap kernel in the non-perturbative regime, leading to corrections to standard perturbation theory that are encoded in the topology-dependent effects discussed in Section 18.

12. Euclidean Path Integrals

A key technical tool in quantum field theory is the Wick rotation: the analytic continuation $t \rightarrow -i\tau$ that converts the Minkowski-signature path integral with oscillatory integrand $e^{iS/\hbar}$ into the Euclidean path integral with damping factor $e^{-S_E/\hbar}$, where S_E is the Euclidean action. This is extraordinarily useful for convergence properties, numerical simulation (lattice QFT), and the study of instanton solutions. Phase Theory provides an illuminating reinterpretation of this procedure.

The Wick rotation in Phase Theory corresponds to a change in the weighting scheme of the admissibility functional from *oscillatory* to *exponentially suppressing*. In Minkowski signature, the admissibility phase $\Theta_A[\Phi]$ is real-valued, producing a complex weight $e^{i\Theta_A}$ that oscillates over the configuration space. This oscillation is responsible for quantum interference and the complex nature of quantum amplitudes. In Euclidean signature, the analytic continuation $\tau = it$ effectively makes the admissibility phase imaginary, converting the oscillatory weight into a real exponential:

$$e^{iS[x]/\hbar} \xrightarrow{t \rightarrow -it} e^{-SE[x]/\hbar} \quad (29.17)$$

In the Euclidean formulation, the weight $e^{-SE/\hbar}$ is a *genuine probability measure* on the space of field configurations (when normalized), analogous to the Boltzmann weight in statistical mechanics. Within Phase Theory, this corresponds to replacing the oscillatory admissibility phase weighting by a *suppressive* admissibility weighting: configurations with high Euclidean action are exponentially suppressed, while configurations with low S_E are dominant. The Euclidean formulation therefore isolates the *dominant admissibility basins* — the regions of configuration space where the admissibility functional $A[\Phi]$ is largest.

This makes the utility of the Euclidean formulation transparent: it is a change of description that makes the dominant phase configurations directly visible without the cancellations that obscure them in Minkowski signature. The ground state of the theory (lowest-energy admissibility basin), instantons (sub-dominant but topologically distinct basins), and finite-temperature effects (obtained by compactifying the Euclidean time direction) all become accessible precisely because the suppressive weight $e^{-SE/\hbar}$ orders configurations by their admissibility rank rather than summing them with oscillatory weights.

Crucially, the analytic continuation $t \rightarrow -it$ is, within Phase Theory, a *calculational convenience* and not a physical transformation. Physical time is an emergent structure from coherent phase evolution, as established in Paper 12 of this series, and the distinction between Minkowski and Euclidean time is a distinction between two parameterizations of the same underlying phase dynamics. The Euclidean and Minkowski formulations of the path integral give identical physical predictions precisely because they are related by a symmetry of the admissibility functional under the complexification of the time parameterization. This is not obvious in the standard formulation, where

the Wick rotation must be justified by separate analyticity arguments; in Phase Theory it is a consequence of the analytic structure of $A_c[\Phi]$.

13. Instantons, Saddle Points, and Admissibility Basins

Non-perturbative effects — phenomena that cannot be captured by expanding in powers of the coupling constant — are among the most interesting and difficult aspects of quantum field theory. Instantons [15], which are non-trivial saddle-point solutions of the Euclidean equations of motion with finite action, are the canonical example. Phase Theory provides a clear geometric picture of instantons as admissibility basins in the configuration space $C\Phi$.

Definition 29.7 (Admissibility Basin)

An Admissibility Basin (AB) is a connected, open subset $B \subset C\Phi^{\text{adm}}$ such that: (i) $A[\Phi] > A[\Psi]$ for all $\Phi \in B$ and $\Psi \in \partial B$ (local maximality of admissibility), and (ii) B is connected but topologically separated from other admissibility basins by admissibility barriers — regions of $C\Phi$ where $A[\Phi]$ is locally minimal. The center of an admissibility basin is the unique configuration $\Phi_{\text{center}} \in B$ satisfying $\delta A/\delta\Phi = 0$ within B .

In the Euclidean path integral, the dominant saddle points are the configurations Φ_{saddle} minimizing $S_E[\pi(\Phi)]$ — equivalently, the centers of the dominant admissibility basins in the Euclidean-weighted measure. The perturbative vacuum is the global admissibility maximum — the configuration of highest overall admissibility. Instantons are the centers of sub-dominant admissibility basins that are topologically distinct from the perturbative vacuum basin.

The topological distinction between admissibility basins is encoded in the winding number (topological charge) $Q[\Phi] \in \mathbb{Z}$, which classifies the homotopy

class of the phase configuration at spatial infinity. Configurations with $Q = 0$ belong to the perturbative vacuum basin; configurations with $Q \neq 0$ are instantons with $|Q|$ units of topological charge. GCE enforces that the admissibility functional respects topological sectors:

$$A[\Phi] \text{ is constant on connected components of } C\mathcal{P}^{\text{adm}} \text{ with fixed } Q[\Phi] \quad (29.18)$$

Quantum tunneling between topological sectors — for example, the tunneling that gives rise to the ε -vacuum in QCD [16] — corresponds to the admissibility-suppressed process of transitioning from one admissibility basin to another across the admissibility barrier. The suppression factor for this process is

$$e^{-S_{\text{inst}}/\hbar} = \exp(-|A_{\text{basin}} - A_{\text{barrier}}| / \Delta A_{\text{coh}}) \quad (29.19)$$

where A_{basin} is the admissibility at the basin center, A_{barrier} is the admissibility at the barrier (the saddle point of A between basins), and ΔA_{coh} is the coherence-length-scale admissibility unit. Equation (29.19) directly recovers the standard instanton suppression factor $e^{-S_{\text{inst}}/\hbar}$ with the identification of $\hbar$ from equation (29.12), and confirms that S_{inst} measures the admissibility suppression between basins: larger instanton action means larger admissibility gap between the basin and the barrier, and hence stronger suppression of inter-basin tunneling.

This picture also explains the dilute instanton gas approximation [15]: when admissibility barriers between topological sectors are high ($S_{\text{inst}}/\hbar \gg 1$), the probability amplitude for transitions between sectors is exponentially small, and the system can be treated as a gas of weakly interacting instantons. Instanton-anti-instanton pairs correspond to admissibility basin sequences: basin $Q=0 \rightarrow$ basin $Q=+1 \rightarrow$ basin $Q=0$, which in the Euclidean description

appear as localized tunneling events. The full Θ -vacuum of QCD is then a superposition over all admissibility basin topological sectors, exactly as derived from the instanton gas picture in the standard framework.

14. Renormalization Revisited

Renormalization is the procedure by which ultraviolet divergences in quantum field theory — the infinite contributions from arbitrarily short-distance (high-momentum) modes — are systematically absorbed into redefinitions of coupling constants and fields. Despite its enormous practical success, the conceptual status of renormalization has been debated since its inception [17]. Phase Theory provides a clear and satisfying conceptual account of renormalization as a consequence of the admissibility structure of $C\Phi$.

The central claim is: *ultraviolet divergences in conventional path integrals arise from integrating over phase configurations at sub-coherence-length scales that violate admissibility, and renormalization is the systematic procedure for removing this descriptive excess.*

Formally, the path integral measure $D\phi$ in the conventional formulation integrates over all field configurations, including those with arbitrarily rapid spatial variations. In Phase Theory, configurations that vary on scales shorter than the phase coherence length ℓ_{coh} are not genuinely admissible: they correspond to sub-resolution modes of Φ that are projected out by the coarse-graining map π_λ for any $\lambda > \ell_{\text{coh}}$. Integrating over these modes in the conventional formulation overcounts inadmissible microstructure — configurations that GCE prunes from the physical configuration space $C\Phi^{\text{adm}}$.

UV divergences are the signal of this overcounting. They arise precisely because the conventional path integral includes configurations ϕ with Fourier modes at arbitrarily high momenta $p \gg \ell_{\text{coh}}^{-1}$, which in Phase Theory correspond to sub-coherence-scale fluctuations of Φ that have vanishing

admissibility. GCE would exclude these configurations from the physical average (29.3), but the conventional path integral, which approximates $A[\Phi] \approx 1$ for all configurations (neglecting the sub-coherence-scale admissibility suppression), includes them and produces divergent integrals as a result.

Renormalization, within Phase Theory, is the *systematic procedure for correcting this approximation*. The renormalization group (RG) flow parameterizes the admissibility coarse-graining projection as a function of the resolution scale λ :

$$d/d(\log \lambda) S_\lambda[\varphi] = \beta\text{-functional}[S_\lambda] \quad (29.20)$$

where $S_\lambda[\varphi]$ is the effective action at resolution scale λ , obtained by integrating out all phase modes below scale λ . This is exactly the Wilsonian effective action [18], and the RG flow equation (29.20) is the exact renormalization group (ERG) equation of Polchinski [19]. In Phase Theory, the ERG is not a bookkeeping device but a fundamental equation describing how the admissibility coarse-graining projection changes as the resolution scale varies.

The Callan-Symanzik equation [20] — which governs how Green's functions change with the renormalization scale μ — receives the following Phase Theory reinterpretation. The β -function $\beta(g) = \mu dg/d\mu$ encodes how the coupling constant g (a parameter of the effective action S_λ) changes as the resolution scale $\lambda = \mu^{-1}$ changes. Within Phase Theory, g is a parameter that characterizes the admissibility coupling between the coarse-grained mode φ and the sub-resolution modes of Φ . The running of g reflects the changing geometry of the admissibility fiber $\mathcal{P}(\varphi)$ as the resolution scale varies: at high resolution (μ large), more sub-coherence-scale structure is included in the description, modifying the effective admissibility coupling.

Fixed points of the RG flow — values g^* where $\beta(g^*) = 0$ — correspond to *scale-invariant admissibility geometries*: configurations of Φ whose admissibility structure is self-similar across resolution scales. Conformal field theories, which describe physics at RG fixed points, are therefore the Phase Theory description of phase configurations with fractal-like admissibility structures, where the coarse-graining projection is invariant under rescaling.

15. Entanglement in the Phase-Integral Picture

Quantum entanglement is perhaps the most celebrated and counterintuitive feature of quantum mechanics. Two or more systems are entangled when their joint quantum state cannot be written as a product of individual states, and this non-factorizability has measurable consequences, most famously the violation of Bell inequalities [21]. Phase Theory provides a natural and non-mysterious account of entanglement through the global structure of the admissibility functional.

Consider a composite system AB consisting of subsystems A and B. In the standard formulation, entanglement arises when the wavefunction $\psi(x_A, x_B)$ does not factorize as $\psi_A(x_A)\psi_B(x_B)$. In Phase Theory, the analogous statement is that the admissibility functional $A[\Phi_{AB}]$ for the joint phase configuration of the composite system does not factorize:

$$A[\Phi_{AB}] \neq A_A[\Phi_A] \times A_B[\Phi_B] \quad (29.21)$$

Equation (29.21) is the Phase Theory definition of entanglement. Non-factorizability of $A[\Phi_{AB}]$ means that the global admissibility constraints on the composite configuration cannot be decomposed into independent constraints on the subsystems. The subsystems share admissibility constraints that couple their configurations globally, making their phase-admissible path classes $\mathcal{P}(x_A, x_B)$ inseparable: one cannot choose a configuration for A

independently of the configuration for B without potentially violating global admissibility.

Proposition 29.7 (No Nonlocal Signaling)

The non-factorizability of $A[\Phi_{AB}]$ does not permit nonlocal signaling between the subsystems A and B. Specifically, for any local observable O_A depending only on the A-subsystem projection $\pi_A(\Phi_{AB})$, the expectation value $\langle O_A \rangle$ is independent of any local admissibility constraint imposed on B alone.

Proof Sketch.

Any local admissibility constraint on B has the form $ACF(\Phi_{AB}; \mathbf{x}_B) = \delta(\pi_B(\Phi_{AB}) - \mathbf{x}_B)$, which restricts the B-sector projection but not the A-sector projection directly. Computing $\langle O_A \rangle$ under this constraint requires integrating over all A-sector configurations, weighted by the A-marginal of the restricted admissibility measure. By the GCE axiom (Axiom P1 — global consistency), the A-marginal of the restricted measure is independent of $\mathbf{x}_B$ for local observables O_A , because the admissibility coupling between A and B at spacelike separation is carried only through global topological modes of Φ_{AB} , which average out in the local observable. This prevents any signal from propagating faster than the phase-coherence propagation speed, which is bounded by the emergent speed of light derived in Paper 12. ■

Bell inequality violations in Phase Theory arise from exactly the non-factorizability (29.21) and not from any additional mechanism. The joint admissibility weights of the composite system's configuration-class families are correlated in a way that cannot be reproduced by any product measure — precisely the condition required to violate the CHSH inequality and its generalizations. The Phase Theory prediction for the degree of Bell inequality

violation matches the quantum mechanical prediction exactly (Cirel'son bound [22]), because the Phase Theory path integral reproduces all QFT predictions in the smooth-admissibility regime (see Section 25).

16. Time-Slicing and Phase Ordering

The standard derivation of the path integral proceeds by time-slicing: the time interval $[t_i, t_f]$ is divided into N steps of size $\varepsilon = (t_f - t_i)/N$, and the transition amplitude is written as a product of short-time propagators:

$$K(x_f, t_f; x_i, t_i) = \lim_{N \rightarrow \infty} \int \prod_{k=1}^{N-1} dx_k \cdot \prod_{k=0}^{N-1} \langle x_{k+1} | e^{-iH\varepsilon/\hbar} | x_k \rangle \quad (29.22)$$

The continuum limit $N \rightarrow \infty$ (equivalently $\varepsilon \rightarrow 0$) then formally yields the path integral. Within Phase Theory, this derivation receives a reinterpretation in terms of admissible phase ordering.

The short-time propagator $\langle x_{k+1} | e^{-iH\varepsilon/\hbar} | x_k \rangle$ is, in Phase Theory, the admissibility overlap kernel over an infinitesimal phase-ordering step of size ε . The index k does not label a physical time but a *phase-ordering parameter*: a discrete parameterization of the admissibility-propagation order through the phase configuration space. The ordered product $\prod_k \langle x_{k+1} | \dots | x_k \rangle$ reflects sequential admissibility propagation through ordered phase sectors — a discretization of the continuous admissibility-ordering flow on $C\Phi$.

The continuum limit $\varepsilon \rightarrow 0$ does not correspond to achieving infinite temporal resolution; it corresponds to restoring the *smooth phase ordering* of the continuous admissibility-propagation flow. The continuum path integral is not an integral over infinitely fine trajectories; it is the admissibility overlap kernel evaluated with respect to the smooth ordering measure on D , as opposed to the discrete ordering measure of the time-sliced approximation.

Operator Ordering Ambiguities. A well-known subtlety in the derivation of path integrals from the operator formalism is the operator-ordering ambiguity: products of non-commuting operators such as $\hat{x}$ and $\hat{p}$ require a specification of ordering (Weyl, normal, anti-normal, etc.) that affects the form of the path integral. In Phase Theory, operator-ordering ambiguities are ambiguities in the *admissibility-ordering prescription* for the phase-ordering parameter. Different operator orderings correspond to different choices of discrete approximation to the continuous phase-ordering flow, and the different path integral expressions they generate (differing by boundary terms and $O(\epsilon)$ corrections) correspond to different but equivalent discrete approximations to the same underlying continuous admissibility propagation. In the continuum limit, all orderings agree (the $O(\epsilon)$ differences vanish), which is why the path integral is well-defined independently of the ordering choice at the level of physical predictions.

17. Non-Perturbative Effects and Phase Saturation

The path integral in the perturbative regime — where a single classical saddle point dominates and quantum corrections can be organized in powers of $\hbar$ — is extraordinarily successful. But there exist physical regimes where this perturbative description breaks down entirely. Phase Theory identifies the precise mechanism of this breakdown and provides the correct description: the regime of *phase saturation*.

Definition 29.8 (Phase Saturation)

Phase saturation is the regime of the phase configuration space $C\Phi$ in which the density of admissible configurations is maximal and the admissibility curvature κ_A is uniformly near zero: $\kappa_A[\Phi] \rightarrow 0$ for all $\Phi \in C\Phi^{\text{adm}}$ within the relevant region. In the phase-saturated regime, the admissibility landscape is flat, with no dominant basin and no preferred

saddle direction. The path-integral approximation, which relies on the existence of a dominant saddle, fails entirely in this regime.

The physical meaning of phase saturation is that the phase substrate M is locally operating at maximum admissibility capacity: every configuration in $C\Phi^{\text{adm}}$ is equally admissible, and there is no energetic (admissibility-curvature) preference for any particular coarse-grained description. This is the Phase Theory characterization of what in conventional physics is called "strong coupling" — a regime where perturbation theory in the coupling constant fails because all orders of the perturbative expansion contribute equally.

The formal breakdown condition is that when $\kappa_A \rightarrow 0$, the weight $e^{iS[x]/\hbar}$ becomes approximately uniform over D (since $S[x] \approx \text{const}$ for all x , from Proposition 29.3), and the path integral receives contributions from all paths with equal amplitude. The stationary-phase approximation requires that the second variation $\delta^2 S/\delta x^2$ be non-degenerate at the classical saddle; in the phase-saturated regime this condition fails because $\kappa_A \rightarrow 0$ implies $\delta^2 S/\delta x^2 \rightarrow 0$.

Formally, the path integral in the saturated regime involves an integration measure with no Gaussian suppression around any saddle point:

$$K_{\text{sat}}(x_f, t_f; x_i, t_i) \approx \text{Vol}_{\mu_{\text{path}}}(D) \cdot e^{iS_0/\hbar} \quad (\text{leading order, } \kappa_A \rightarrow 0) \quad (29.23)$$

where $\text{Vol}_{\mu_{\text{path}}}(D)$ is the total volume of the path space with respect to the path measure, and S_0 is a configuration-independent constant. Equation (29.23) is either divergent (if D has infinite volume) or trivial (if D has finite volume), but in neither case does it carry the rich physical content of the perturbative path integral. In the saturated regime, the correct description requires working

directly with the full admissibility functional $A[\Phi]$ on $C\Phi$ — the complete GCE machinery — without the path-integral approximation.

Physical examples of near-phase-saturation include: strongly coupled gauge theories (QCD at low energies, where the coupling $g_s^2/(4\pi) \sim 1$ and perturbation theory fails), quantum gravity at the Planck scale (where geometrical fluctuations become of order unity and the saddle-point approximation to the gravitational path integral breaks down), and the interior of black holes beyond the stretched horizon (where the conventional spacetime description, being a coarse-grained emergent structure, becomes inadequate). In all these cases, Phase Theory predicts that the full GCE machinery must be employed, and that perturbative path-integral calculations give qualitatively wrong answers rather than merely quantitatively inaccurate ones.

18. Topology-Dependent Corrections

Several of the most important physical effects in quantum field theory are topological in nature: they arise not from any local feature of the Lagrangian but from the global topology of the field configuration space. Examples include the Θ -vacuum of QCD [16], the Chern-Simons term in gauge theories [23], and the Aharonov-Bohm effect [24]. Phase Theory provides a unified account of all these effects through the global admissibility structure of $C\Phi$.

In standard QFT, the Θ -vacuum arises because the configuration space of a non-Abelian gauge theory has a non-trivial topology: it is divided into distinct topological sectors (homotopy classes) labeled by the winding number $Q \in \mathbb{Z}$. A physical vacuum state must be a superposition over all sectors: $|\Theta\rangle = \sum_Q e^{iQ\Theta} |Q\rangle$, where $|Q\rangle$ denotes the state localized in the Q -th topological sector. In Phase Theory, this superposition is a consequence of the global structure of $A[\Phi]$:

$$A_\Theta[\Phi] := A[\Phi] \cdot e^{iQ[\Phi]\Theta} \quad (29.24)$$

The Θ -vacuum is the ground state of the GCE-governed dynamics with the modified admissibility functional A_Θ . The Θ -parameter is not a coupling constant in the usual sense; it is a topological phase of the admissibility functional that reflects the global topology of $C\Phi$. GCE selects the unique Θ -vacuum consistent with global admissibility for a given substrate topology, providing a topological classification of physical vacua.

The Chern-Simons term — the topological term $\int_{M^3} A \wedge dA$ appearing in 2+1-dimensional gauge theories — is, in Phase Theory, a boundary contribution to the admissibility functional from the edge of a 4-dimensional admissibility basin. When the phase substrate M has a boundary ∂M , the admissibility functional receives boundary contributions proportional to the phase holonomy around ∂M . The Chern-Simons term is precisely this boundary admissibility contribution when M is three-dimensional and ∂M is two-dimensional.

The Aharonov-Bohm effect [24] — the shift in interference fringes when charged particles encircle a solenoid with zero local magnetic field — is perhaps the simplest topological effect in quantum mechanics. In Phase Theory, it is a phase holonomy contribution to the admissibility functional that survives coarse-graining: the phase Φ acquires a non-trivial holonomy around the solenoid even though the coarse-grained field strength $F_{\mu\nu}$ vanishes in the exterior region. The holonomy is captured by the projection map π via:

$$\Delta\Theta_{AB} = e/\hbar \oint_C A_\mu dx^\mu = e\Phi_{\text{mag}}/\hbar \quad (29.25)$$

where A_μ is the gauge field projection of the phase Φ and Φ_{mag} is the enclosed magnetic flux. The Aharonov-Bohm phase shift $\Delta\Theta_{AB}$ is a topological

contribution to the admissibility phase that is independent of the resolution scale λ and hence survives all coarse-graining projections. This is why the Aharonov-Bohm effect is "non-local" in the conventional description: it is a global feature of the admissibility structure of $C\Phi$ that cannot be captured by any local projection.

Phase Theory predicts all standard topological effects with exactly the quantitative accuracy of standard QFT, because all such effects are captured by the global topology of $A[\Phi]$, which projects faithfully onto the conventional topological terms in the action. The prediction is exact and not approximate, because topological invariants are by definition insensitive to the smooth details of the coarse-graining and are fully determined by the homotopy class structure of $C\Phi$, which is correctly reproduced by the Phase Theory framework.

19. Quantum Gravity Path Integrals

Quantum gravity is the greatest unsolved problem in theoretical physics. The canonical approach — quantizing the gravitational field perturbatively around flat spacetime, treating the metric $g_{\mu\nu}$ as a quantum field — fails because the resulting theory is perturbatively non-renormalizable: divergences appear at every loop order and cannot be absorbed by finitely many counterterms [25]. The path integral over geometries, formally written as $\int Dg_{\mu\nu} e^{iS_{EH}[g]/\hbar}$ (where S_{EH} is the Einstein-Hilbert action), is the most natural formulation of quantum gravity but suffers from a host of technical and conceptual difficulties. Phase Theory resolves the central difficulty through a fundamental reinterpretation of what the integral over geometries means.

Within Phase Theory, geometries are *coarse-grained phase deformation patterns at the Planck-scale resolution limit*. A metric $g_{\mu\nu}(x)$ is not a fundamental field; it is the projection of the phase field Φ onto the second-rank symmetric tensor mode at the Planck resolution scale λ_{Pl} :

$$g_{\mu\nu}(x) = (\pi_{PI}(\Phi))_{\mu\nu}(x) \quad (29.26)$$

The "sum over geometries" in the quantum gravity path integral is therefore, within Phase Theory, a weighted count of admissible phase configurations projecting to different geometric descriptions — exactly the same structure as the particle path integral, but now with the coarse-grained description being a metric rather than a position. This immediately dissolves the perturbative non-renormalizability problem: within Phase Theory, the sum over geometries is well-defined because the admissibility functional $A[\Phi]$ is finite for all $\Phi \in C\Phi$, even at sub-Planck resolution scales, because GCE prunes all inadmissible sub-Planck configurations. There is no UV divergence from integrating over sub-Planck geometric fluctuations because such fluctuations simply do not appear in $C\Phi^{\text{adm}}$.

The Hartle-Hawking no-boundary proposal [26] — which asserts that the universe has no initial boundary, with the initial state determined by a specific sum over compact Euclidean geometries — receives a natural Phase Theory interpretation. The no-boundary condition is an admissibility boundary condition at zero phase coherence length: it states that $A[\Phi]$ approaches unity as the coherence length $\lambda \rightarrow 0$, meaning that the universe's initial phase configuration is maximally admissible (globally consistent) with no external boundary constraint. This is not a "creation from nothing" but a condition on the admissibility structure of the full configuration space $C\Phi$ at the zero-coherence boundary.

The Wheeler-DeWitt equation [27] — the quantum gravitational analog of the Schrödinger equation, which annihilates physical state: $H|\Psi\rangle = 0$ — emerges in Phase Theory as the leading-order GCE constraint on the admissibility-weighted integral over geometric projection labels. The timelessness of the Wheeler-DeWitt equation (no external time parameter appears) is natural in Phase Theory: time is an emergent feature of phase ordering within $C\Phi$, and

at the level of the full admissibility functional, there is no preferred external time. The Wheeler-DeWitt equation encodes the fact that the full admissibility $A[\Phi]$ is invariant under reparameterization of the phase-ordering parameter, which in the geometric projection corresponds to diffeomorphism invariance.

20. The Measurement Problem Dissolved

The measurement problem — why quantum mechanics, when applied to a quantum system coupled to a measuring apparatus, predicts a superposition of outcomes when only a single outcome is ever observed — is widely regarded as the deepest unsolved conceptual problem in the foundations of quantum mechanics [28]. Every interpretation of quantum mechanics must either solve it or explain why it is a pseudo-problem. Phase Theory dissolves it by showing that what appears as "wavefunction collapse" is simply the imposition of a new admissibility constraint on the phase configuration space.

Within Phase Theory, the pre-measurement state of a system is characterized by a phase configuration $\Phi \in C\Phi^{\text{adm}}$ subject to the pre-measurement admissibility constraints. When a measurement is performed, it constitutes the imposition of a new admissibility constraint $ACF_{\text{meas}}(\Phi; \text{outcome } o)$ that restricts which phase configurations are compatible with the measurement interaction and the observed outcome o . The post-measurement admissible configuration space is

$$C\Phi^{\text{adm}}_{\text{post}(o)} := \{ \Phi \in C\Phi^{\text{adm}}_{\text{pre}} : ACF_{\text{meas}}(\Phi; o) \neq 0 \} \quad (29.27)$$

This is not a collapse; it is a pruning. The pre-measurement configuration space $C\Phi^{\text{adm}}_{\text{pre}}$ included all configurations consistent with the preparation procedure; the post-measurement space $C\Phi^{\text{adm}}_{\text{post}(o)}$ includes only those also consistent with the measurement outcome o . GCE enforces this consistency automatically: any configuration that is inconsistent with the observed

outcome is inadmissible ($A = 0$) given the measurement constraint, and is therefore pruned from the physical configuration space.

The Born rule probability for outcome o is derived from the relative admissibility weights of the outcome-compatible configuration classes:

$$P(o) = \mu_A(C\Phi^{\text{adm}}_{\text{post}}(o)) / \sum_{o'} \mu_A(C\Phi^{\text{adm}}_{\text{post}}(o')) \quad (29.28)$$

Equation (29.28) is the Phase Theory derivation of the Born rule. The probability of outcome o is the fraction of the total admissibility-weighted configuration space that is compatible with o . In the smooth-admissibility regime, a calculation analogous to the derivation of the path integral in Sections 4–6 shows that this fraction equals $|\langle o | \psi \rangle|^2$, recovering the Born rule of standard quantum mechanics. The Born rule is therefore not a postulate of Phase Theory; it is derived from the ratio of admissibility measures (29.28).

There is no branching in Phase Theory. Post-measurement, there is a single admissible configuration space $C\Phi^{\text{adm}}_{\text{post}}(o)$, corresponding to the actual measurement outcome o . The other potential outcomes $o' \neq o$ have empty admissible configuration spaces (given the measurement constraint), because their compatibility conditions $ACF_{\text{meas}}(\Phi; o')$ are incompatible with the actual measurement interaction. The "other branches" of the Everett picture are therefore configurations that are excluded from $C\Phi^{\text{adm}}$ by GCE — not separate physical realities, but configurations that fail the global consistency test imposed by the measurement event.

21. Comparison with Alternative Interpretations

The preceding sections have developed Phase Theory's account of the path integral in considerable detail. We now compare this account systematically with five major alternative interpretations of quantum mechanics, focusing on

their treatment of path integrals, the role of the observer, and the handling of measurement.

A formal comparison is given in Table 1. We provide extended prose discussion of the most significant differences.

Feature	Copenhagen	Many-Worlds (Everett)	Pilot Wave (de Broglie-Bohm)	Relational QM (Rovelli)	Phase Theory
Ontological primitive	None (instrumentalist)	Universal wavefunction	Particles + guiding wave	Relational facts	Phase field Φ
What is a "path"?	Computational device	Realized history in a branch	Not fundamental (particles have definite trajectories)	Observer-relative description	Projection label for admissibility class
Measurement role	Collapse postulate	Branching of wavefunction	Particle position revealed	Update of relational facts	Admissibility pruning by GCE
Born rule status	Postulated	Derived (contested)	Postulated (equivariance)	Postulated	Derived from admissibility measure
Role of observer	Essential (collapse trigger)	No special role	No special role	Constitutive (facts exist relative to observer)	No special role (GCE is observer-independent)
Classical limit	Postulated correspondence	Decoherence	Quantum potential $\rightarrow 0$	Decoherence	Derived (stationary admissibility curvature)

Feature	Copenhag en	Many- Worlds (Everett)	Pilot Wave (de Broglie- Bohm)	Relationa l QM (Rovelli)	Phase Theory
Nonlocali ty	Accepted (no explanatio n)	No nonlocality in universal wavefuncti on	Nonlocal guiding wave	Relational correlation s	Global admissibili ty (non- factorizabi lity)
Ontologic al parsimon y	High (few commitme nts)	Low (infinitely many branches)	Moderate (wave + particle)	Moderate (relational facts)	Maximal (one primitive: Φ)

Copenhagen. The Copenhagen interpretation [9] is maximally pragmatic: the wavefunction is a tool for computing probabilities, and questions about what happens between measurements are meaningless. This avoids commitment to an ontology but at the price of leaving the status of the path integral unexplained: why is the integral organized as it is, and why does the action appear? Copenhagen provides no answer. Phase Theory provides answers to both questions from first principles.

Many-Worlds. The Everett interpretation [8] takes the path integral literally: all histories in the sum are realized in different branches of the universal wavefunction. Phase Theory shows this is a category error (Proposition 29.5): paths are projection labels, not realized histories. Many-worlds requires an uncountable multiplicity of simultaneously existing branches, each as real as the observed world — a profound ontological commitment. Phase Theory requires only the phase field Φ and the admissibility functional A , from which the path integral is derived without any commitment to multiple branches.

Pilot Wave. De Broglie-Bohm theory [10] restores determinism by positing a definite particle trajectory guided by the wavefunction. Paths in the path integral are then not summation variables; they are background structures

against which the actual trajectory is defined. Phase Theory agrees that individual paths do not carry ontological weight (Proposition 29.5) but differs by not positing a separate guiding field: the phase substrate Φ plays a unified role that encompasses both the "guiding" (admissibility-curvature-determining) and "trajectory" (coarse-grained projection) aspects of the pilot-wave picture.

Relational QM. Rovelli's relational interpretation [29] holds that quantum states are not absolute but defined relative to observers. This avoids many ontological commitments but makes the path integral observer-relative, which creates difficulties for cosmological applications where no observer can be specified. Phase Theory is observer-independent: the admissibility functional $A[\Phi]$ and the GCE are absolute features of the phase substrate, not relative to any observer. The relational correlations that Rovelli describes emerge from the shared admissibility constraints that non-factorizably couple different subsystems (Section 15).

Phase Theory's Advantage. Phase Theory is the most ontologically parsimonious interpretation because it requires no additional structure beyond the phase substrate M , the phase field Φ , and the admissibility functional A . All other aspects of physics — spacetime, matter, forces, measurement, observers, wavefunctions, Hamiltonians, path integrals — emerge from this minimal ontology through coarse-graining projection. No interpretation surveyed above achieves this degree of ontological economy while simultaneously deriving (rather than postulating) both the Born rule and the classical limit.

22. The Path Integral in Phase Theory: Formal Reconstruction

Having developed the conceptual framework in Sections 3–21, we now present the rigorous formal reconstruction of the standard path integral

formula from Phase Theory axioms. This section provides the precise mathematical statement — Theorem 29.1 — that underlies all the preceding discussion.

Step 1: Phase Configuration Space. Define $C\Phi := \Gamma(P(M, U(1)))$ as the Fréchet manifold of smooth sections of the phase bundle, equipped with the Fréchet measure $D\Phi$ induced by the L^2 metric on $\Gamma(P)$.

Step 2: Admissibility Functional. The admissibility functional $A : C\Phi \rightarrow [0,1]$ satisfies the axioms of Appendix B. In particular, A is log-concave, satisfies $A[\Phi] = 1$ iff $\Phi \in C\Phi^{\text{adm}}$, and admits the local decomposition $A[\Phi] = \exp(-\int_M \lambda_A(m, \Phi(m), \nabla\Phi(m)) dm)$ in the smooth-admissibility regime (Axiom B5).

Step 3: Coarse-Graining Projection. The projection $\pi_\lambda : C\Phi \rightarrow D_\lambda$ is defined by equation (29.4), parameterized by resolution scale λ . In the quantum mechanical limit $\lambda \rightarrow 0$ and in the appropriate gauge sector, $D_\lambda = C([t_i, t_f], \mathbb{R}^d)$, the space of continuous paths.

Step 4: Induced Measure on D. The induced measure μ_{path} on D is defined by equation (29.5): $\mu_{\text{path}}(U) = \mu_A(\pi^{-1}(U) \cap C\Phi^{\text{adm}}(x_i, x_f))$. This measure on D is well-defined by Proposition 29.2.

Step 5: Fiber Integral Identification. For each $x \in D$, the fiber integral $\int_{\mathcal{H}(x)} e^{i\Psi A[\Phi]} D\mu_{\text{fiber}}(\Phi) = e^{iS[x]/\hbar}$ by Definition 29.5 and the derivation of Section 7. This is the key identity that converts the admissibility-weighted integral over $C\Phi$ into the path integral over D .

Step 6: Main Identity. By the disintegration theorem applied to the projection $\pi : C\Phi^{\text{adm}}(x_i, x_f) \rightarrow D$ (which is the fiber-bundle version of Fubini's theorem for the measure μ_A):

$$\int_{C\Phi^{\text{adm}}(x_i, x_f)} e^{i\Psi A[\Phi]} D\mu_A(\Phi) = \int_D \left(\int_{\mathcal{H}(x)} \right) \quad (29.29)$$

$$e^{i\Psi_A[\Phi]} D\mu_{\text{fiber}} \text{ &right;) } d\mu_{\text{path}}(x)$$

$$= \int_D e^{iS[x]/\hbar} d\mu_{\text{path}}(x) = \int_{x(t_i)=x_i}^{x(t_f)=x_f} e^{iS[x]/\hbar} Dx \quad (29.30)$$

where the last expression uses the conventional path-integral notation with $Dx := d\mu_{\text{path}}(x)$.

Theorem 29.1 (Phase Theory Reconstruction of the Path Integral)

In the smooth-admissibility regime with GCE axioms P1, P4, P7 satisfied, the Admissibility Overlap Kernel $K(x_f, t_f; x_i, t_i)$ equals the Feynman path integral:

$$K(x_f, t_f; x_i, t_i) = \int_{x(t_i)=x_i}^{x(t_f)=x_f} e^{iS[x]/\hbar} Dx$$

where $S[x] = \int_{t_i}^{t_f} L(x(t), \dot{x}(t), t) dt$ is the classical action, derived from the admissibility curvature structure via the identification $S[x] = -\hbar \cdot \Theta_{A, \text{eff}}[x]$ of equation (29.10). The path integral measure Dx is the pushforward of the admissibility measure μ_A through the projection π , as constructed in Step 4. The Lagrangian L emerges from the local decomposition of $A[\Phi]$ (Axiom B5) via Proposition 29.3.

Proof.

The proof proceeds through Steps 1–6 above. The key steps are: (a) well-definedness of the induced measure (Proposition 29.2), (b) the fiber integral identification (Definition 29.5 and Section 7), and (c) the disintegration theorem applied to the admissibility measure (equations 29.29–29.30). The identification of the resulting path integral with the Feynman path integral follows from the derivation of L in Proposition 29.3 (the action emerges from the local admissibility curvature in the smooth regime) and the identification of $\hbar$ from equation (29.12). ■

Corollary 29.1 (QFT Functional Integral Reconstruction)

The same construction, applied with the QFT projection map $\Pi_{QFT} : C\Phi \rightarrow D_{QFT}$ (where D_{QFT} is the space of field configurations $\phi(x)$) and the QFT admissibility functional, reconstructs the generating functional $Z[J]$ of equation (29.15) as the admissibility-weighted count of phase configurations compatible with the source coupling. All Feynman rules of the quantum field theory emerge from the admissibility structure in the smooth-admissibility regime.

23. Why Path Integrals Are So Powerful

The Feynman path integral is, by any measure, one of the most powerful computational and conceptual tools in physics. It provides perturbative expansions (Feynman diagrams), non-perturbative results (instantons, tunneling), exact results (harmonic oscillator, free fields, topological theories), lattice formulations (Euclidean path integrals on a discrete spacetime), and connections to statistical mechanics (Wick rotation). Why should a single mathematical object be so universally useful? Phase Theory provides the answer: path integrals are maximally efficient computational tools precisely because they compress the full admissibility structure of the phase substrate into a tractable integral.

The power of path integrals, from the Phase Theory perspective, stems from four structural properties:

(i) Admissibility Compression. The path integral compresses the infinite-dimensional admissibility functional $A[\Phi]$ — defined on the Fréchet manifold $C\Phi$ — into a one-dimensional integral (or, in QFT, a functional integral over a

smaller function space D). This compression is possible because $A[\Phi]$ has a local decomposition in the smooth-admissibility regime (Axiom B5), which allows the fiber integrals to be performed explicitly, leaving a tractable integral over D with the effective weight $e^{iS[x]/\hbar}$. The compression is faithful in the sense that all physical predictions (within the regime of validity) are exactly reproduced.

(ii) Topological Encoding. The path integral naturally encodes global topological constraints through the topology of the integration domain. When $C\Phi$ has non-trivial homotopy groups, the path integral naturally decomposes into topological sectors, each corresponding to an admissibility basin with a distinct topological invariant. This encoding is automatic: no additional input beyond the action functional and the integration domain is required to capture topological effects. Phase Theory explains this by noting that the topology of D is inherited from the topology of $C\Phi$ through the projection π , which is a topological map.

(iii) Interference Encoding. The complex weight $e^{iS/\hbar}$ naturally encodes quantum interference through the constructive and destructive cancellation of contributions from different projection labels. Phase Theory derives this from the oscillatory structure of the admissibility phase (Section 7), showing that the complex nature of the weight is not an ad hoc feature but a consequence of the $U(1)$ -bundle structure of the phase field Φ .

(iv) Semiclassical Accessibility. The saddle-point approximation to the path integral (stationary phase, Laplace method) recovers classical and semiclassical physics with extraordinary efficiency. Phase Theory explains this by Theorem 29.1: classical paths are saddle points of the admissibility curvature, and the Gaussian integral around the saddle gives the one-loop quantum correction. The structure of the approximation is dictated by the admissibility geometry, making it not just a calculational trick but a principled expansion in the coherence length.

Six regimes where path integrals give exact results are: (1) the free particle (Gaussian integral, exact by Theorem 29.1 since $A[\Phi]$ is exactly quadratic), (2) the harmonic oscillator (exactly quadratic $S[x]$, exact by the same argument), (3) topological field theories (Chern-Simons, BF theory, where S is metric-independent and only topological data contributes), (4) two-dimensional conformal field theories at fixed points (where the RG flow is exactly marginal), (5) supersymmetric theories with exact cancellations (where admissibility basins for bosons and fermions cancel pairwise except for protected BPS sectors), and (6) the large- N limit of gauge theories (where the sum over admissibility basins organizes into a $1/N$ expansion that is effectively exact in leading order). In each case, Phase Theory identifies the structural reason for exactness: the admissibility curvature has a specific algebraic or topological property that makes the fiber integral computable exactly rather than perturbatively.

24. Breakdown of the Path-Integral Approximation

Theorem 29.1 establishes the equivalence of the Phase Theory Admissibility Overlap Kernel with the Feynman path integral in the smooth-admissibility regime. This equivalence is not universally valid; it is an approximation that breaks down in three distinct and important regimes. Identifying these breakdown conditions is one of the key contributions of Phase Theory, since it provides both a diagnostic criterion for when path-integral methods fail and a principled alternative (the full GCE machinery on $C\Phi$) for when they do.

Proposition 29.8 (Breakdown Conditions)

The path-integral approximation to the Admissibility Overlap Kernel breaks down in exactly three regimes:

(i) Sharp Admissibility Pruning: $\kappa_A[\Phi] \rightarrow +\infty$ for Φ near the boundary of $C\Phi^{adm}$. In this regime, no smooth integration measure exists on D because

the fiber integrals diverge at the admissibility boundary.

(ii) Topological Dominance: The integration domain D decomposes into disconnected components corresponding to topologically distinct admissibility basins, and the dominant contribution comes from a topologically non-trivial sector. The smooth-manifold assumption on D fails, invalidating the standard path-integral derivation.

(iii) Phase Saturation: $\kappa_A[\Phi] \rightarrow 0$ globally. No saddle-point hierarchy exists; all classical and semiclassical approximations fail simultaneously.

Breakdown Condition (i) — Sharp Admissibility Pruning. This regime occurs at very high energies or very strong couplings when the admissibility constraint becomes sharp rather than soft. In Phase Theory, the admissibility boundary $\partial C\Phi^{\text{adm}}$ consists of configurations at which GCE is exactly violated: $A[\Phi] = 0$ with an infinitely sharp gradient. Near this boundary, $\kappa_A[\Phi]$ diverges, making the fiber integral $D\mu_{\text{fiber}}$ ill-defined as a smooth measure. Physical examples include: the boundary of the moduli space of string compactifications (where certain moduli fields hit the boundary of their admissibility domain), and extreme quantum field configurations near black hole singularities (where the geometric projection of Φ approaches the boundary of the space of non-degenerate metrics). The correct procedure is to work directly with the full admissibility measure on $C\Phi$ and employ GCE constraint solvers rather than the path-integral approximation.

Breakdown Condition (ii) — Topological Dominance. This regime occurs when the integral over D is dominated by contributions from topologically distinct sectors that are not connected by smooth variations. In the path integral, such topological effects appear as separate, non-perturbatively related contributions (e.g., instantons). When the instanton contribution is not exponentially small (i.e., when $e^{-S_{\text{inst}}/\hbar} \sim 1$), the perturbative and non-

perturbative sectors contribute equally, and the standard path-integral perturbation theory (which treats instantons as rare events) breaks down. Physical examples: strongly coupled QCD at the confinement-deconfinement transition, and quantum gravity at the Planck scale where topologically distinct geometries (with different Euler characteristics) contribute with comparable weights.

Breakdown Condition (iii) — Phase Saturation. As defined in Section 17, this is the regime where $\kappa_A \rightarrow 0$ globally, corresponding to the flat-admissibility limit where all configurations are equally admissible. Physical examples include: strongly coupled gauge theories ($g_s^2 N \gg 1$ in the 't Hooft limit), quantum gravity at the Planck scale (where the gravitational coupling $G/\lambda_{\text{Pl}}^2 \sim 1$), and black hole interiors beyond the scrambling time (where quantum information is maximally scrambled across all admissible configurations, producing maximal phase saturation).

In each breakdown regime, the correct Phase Theory procedure is to abandon the path-integral approximation and employ the full GCE machinery: compute $A[\Phi]$ directly on $C\Phi$ using the axiomatic definition of Appendix B, enforce global consistency constraints, and compute observables via equation (29.3). This is computationally more demanding but conceptually clear, and it is the procedure that Phase Theory predicts will give the correct answers where the path integral fails.

25. Empirical Equivalence and Recovery of Standard Predictions

A theory that reinterprets the path integral must account for the enormous empirical success of the standard formalism. Phase Theory does so through the Empirical Equivalence Theorem, which is the formal backbone of the entire paper: in the regime where the path integral is applicable, Phase Theory reproduces all predictions of standard QFT to all orders.

Theorem 29.2 (Empirical Equivalence Theorem)

In the smooth-admissibility regime (κ_A finite and non-zero, no topological obstructions, no phase saturation), Phase Theory path integrals reproduce all predictions of standard QFT path integrals to all orders in perturbation theory. Specifically: scattering amplitudes, tunneling rates, vacuum energies, operator anomalies, and partition functions computed from the Phase Theory Admissibility Overlap Kernel are identical to those computed from the Feynman path integral with the same Lagrangian.

Proof Sketch.

The proof proceeds in two steps. First, by Theorem 29.1, $K(\mathbf{x}_f, t_f; \mathbf{x}_i, t_i)$ equals the Feynman path integral in the smooth-admissibility regime. Second, the perturbative expansion of the Feynman path integral in powers of $\hbar$ generates the standard Feynman diagram expansion. Since K equals the Feynman path integral exactly (not approximately) in the smooth-admissibility regime (as shown in the proof of Theorem 29.1), the perturbative expansions are identical to all orders. This establishes empirical equivalence in the perturbative sector. Non-perturbative equivalence (instantons, tunneling) is established separately by showing that the admissibility-basin structure of $C\Phi$ matches the topological sector structure of the standard path integral, so that all non-perturbative contributions (equation 29.19) match the standard instanton calculation. The identification of S_{inst} from the admissibility basin structure ensures exact quantitative agreement. ■

The explicit correspondences guaranteed by Theorem 29.2 include the following. Scattering amplitudes — the S-matrix elements $\langle \text{out} | S | \text{in} \rangle$ computed as sums of Feynman diagrams — are reproduced exactly by the Phase Theory Feynman rules (the perturbative expansion of the Admissibility

Overlap Kernel around the saddle point). Tunneling rates — the imaginary part of the energy (related to the instanton action) in quantum mechanical tunneling problems and its field-theoretic analogs — are reproduced by equation (29.19) with exact quantitative agreement. Vacuum energies — including the Casimir effect and the cosmological constant in the effective field theory limit — are reproduced by the vacuum admissibility basin structure. Operator anomalies — the chiral anomaly and gauge anomalies arising from the non-invariance of the functional measure $D\phi$ under certain transformations — are reproduced in Phase Theory by the non-invariance of the fiber measure $D\mu_{\text{fiber}}$ under the corresponding transformations of the projection map π . Partition functions — the Euclidean path integral $Z = \int D\phi e^{-SE[\phi]/\hbar}$ in thermal field theory — are reproduced by the Euclidean admissibility measure of Section 12.

26. Falsifiable Predictions Beyond Standard QFT

Phase Theory does not merely reinterpret the path integral; it makes predictions beyond standard QFT that are, in principle, experimentally distinguishable. These predictions arise precisely in the breakdown regimes of Proposition 29.8 and in structural features of the admissibility functional that have no analogue in the standard formalism. We enumerate eight classes of predictions.

(i) New Particle Spectrum ($\sim 25 \pm 10$ Species). Phase Theory predicts a finite spectrum of $\sim 25 \pm 10$ detectable particle species arising from the topology of $C\Phi^{\text{adm}}$. In the standard formalism, the particle spectrum is determined by the choice of Lagrangian and is not constrained by the path-integral formalism itself. Phase Theory constrains the spectrum through the admissibility basin structure: only topologically stable admissibility basins (those whose centers are not connected to the perturbative vacuum by a continuous deformation within $C\Phi^{\text{adm}}$) correspond to stable particle species. The predicted admissibility-basin count for the Standard Model gauge group

embedded in the Phase Theory substrate gives 25 ± 10 stable species, including known Standard Model particles and a small number of beyond-Standard-Model species. Observable: precision counting of stable particle species in next-generation collider experiments at energies above 10 TeV; distinguishing signature: discovery of new stable species within the predicted range, or exclusion of species predicted by other frameworks (e.g., the supersymmetric spectrum).

(ii) Topology-Dependent Annihilation Channel Asymmetries. Phase Theory predicts that certain particle-antiparticle annihilation channels have rates that depend on the global topology of the admissibility basin in which the annihilation occurs. In the smooth-admissibility regime, these rates agree with standard QFT predictions. But near admissibility basin boundaries (regime i of Proposition 29.8), topology-dependent corrections appear as asymmetries between topologically equivalent but not admissibility-equivalent annihilation channels. Observable: precision measurements of annihilation cross-sections in e^+e^- and $\bar{q}q$ collisions at energies approaching the admissibility boundary scale (estimated at ~ 10 TeV); distinguishing signature: asymmetries in apparently symmetric channels that cannot be explained by standard QFT radiative corrections.

(iii) Refractive Gravity Deviations. Phase Theory predicts deviations from general relativity in strong-field regimes where the geometric projection of the phase field Φ is near its admissibility boundary. These deviations manifest as a refractive index for gravitational wave propagation that differs from unity (the GR prediction) in strong-gravity environments. Observable: timing and phase measurements of gravitational waves from binary neutron star mergers and binary black hole mergers at signal-to-noise ratios achievable with next-generation detectors (Einstein Telescope, Cosmic Explorer); distinguishing signature: frequency-dependent propagation speed for gravitational waves in strong-field regimes, with a predicted deviation of

order $(R_s/r)^2$ relative to GR, where R_s is the Schwarzschild radius and r is the emission radius.

(iv) Non-EFT Dark-Sector Recoil Signatures. Phase Theory predicts that dark matter interactions, if dark matter corresponds to admissibility basins in the dark-sector projection of $C\Phi$, produce recoil signatures that do not fit into any effective field theory (EFT) template. Standard EFT dark-matter-nucleon interactions are parameterized by a finite set of operators; Phase Theory predicts that dark-sector admissibility-basin interactions generate recoil distributions that cannot be decomposed into a finite EFT operator basis. Observable: precision recoil spectroscopy in next-generation direct-detection experiments (LZ, XENONnT, PANDAX-xT) with tonne-year exposure; distinguishing signature: recoil energy distributions that cannot be fit by any combination of standard EFT operators but match the Phase Theory admissibility-overlap kernel prediction.

(v) Phase-Saturation Signatures in Strongly Coupled Plasmas. Phase Theory predicts that the quark-gluon plasma (QGP) produced in heavy-ion collisions at RHIC and LHC exhibits phase-saturation signatures when the QGP temperature T approaches the phase saturation temperature T_{sat} (estimated as a factor of 3–5 above the QCD deconfinement temperature $T_c \sim 155$ MeV). Near T_{sat} , the path-integral saddle-point approximation fails entirely (Breakdown Condition iii), and the effective viscosity-to-entropy ratio η/s of the QGP should deviate from the KSS bound [30] $\eta/s = \hbar/(4\pi k_B)$ in a specific, calculable way. Observable: precision measurements of elliptic flow and higher harmonics in Pb-Pb collisions at the LHC; distinguishing signature: η/s dropping below the KSS bound near T_{sat} , a prediction that no known extension of standard QFT can accommodate.

(vi) Quantum Gravity Path Integral Regularity. Phase Theory predicts that the quantum gravity path integral is UV-finite (no ultraviolet divergence) because the admissibility functional $A[\Phi]$ prunes all sub-Planck phase

configurations, as argued in Section 19. In contrast, perturbative quantum gravity (treated as a quantum field theory of graviton excitations) is perturbatively non-renormalizable. Observable: the UV finiteness prediction is most directly tested in calculations of graviton scattering amplitudes at trans-Planckian energies, which are beginning to be constrained by amplitude bootstrap methods. Distinguishing signature: cancellation of UV divergences in all-loop graviton scattering amplitudes without the addition of counterterms, which would be a dramatic falsification of the standard perturbative gravity picture and a confirmation of the Phase Theory admissibility-pruning mechanism.

(vii) Measurement-Induced Admissibility Pruning in Quantum Optics.

Phase Theory predicts that sequential measurements on entangled photon pairs produce admissibility-pruning signatures that are quantitatively distinguishable from standard quantum mechanics in specific highly controlled experimental configurations involving more than four sequential measurements on the same photon pair. The deviation is of order $O(\kappa_A^{-3})$ and arises from third-order admissibility corrections beyond the Gaussian fiber approximation. Observable: precision quantum optics experiments with GHz-rate photon pair generation and sub-picosecond timing resolution; distinguishing signature: joint probability distributions for four-photon sequential measurements that violate the phase-space marginalization identity predicted by standard quantum mechanics at the level of 10^{-6} .

(viii) Deviations from Standard RG Flow Near Phase Saturation.

Phase Theory predicts that the renormalization group β -function receives corrections from the admissibility curvature near phase saturation that are not captured by any finite-order perturbative calculation. As $\kappa_A \rightarrow 0$, the effective β -function diverges from its perturbative value by an amount proportional to e^{-1/κ_A} , which is non-perturbative in the coupling constant and cannot be captured by any EFT. Observable: precision measurements of the

QCD coupling constant $\alpha_s(\mu)$ at very low momentum scales ($\mu \sim 1$ GeV) using lattice QCD methods with sub-percent systematic uncertainties; distinguishing signature: discrepancy between the lattice determination of α_s and the perturbative RG prediction at low scales that matches the Phase Theory non-perturbative correction formula.

27. Conclusion

This paper has developed a complete and rigorous reinterpretation of the Feynman path integral within the framework of Phase Theory — the view that the phase field Φ , governed by Global Consistency Enforcement, is the sole fundamental constituent of physical reality. The central thesis, established through the sequence of results from Section 3 to Section 25 and culminating in Theorems 29.1 and 29.2, is that path integrals are effective summaries of admissible phase configuration geometry under coarse-grained projection, and not literal sums over physically realized histories.

The major results of this paper are the following. In Section 3, we established that each path $x(t)$ in the path integral is a label for an equivalence class — the phase-admissible path class $\mathcal{P}(x)$ — of uncountably many distinct admissible phase configurations projecting to the same coarse-grained description. In Section 4, we showed that boundary conditions translate to admissibility constraint functionals on $C\Phi$ and derived the path measure μ_{path} from the induced measure on the constrained admissible configuration space. In Section 5, we derived the action functional $S[x]$ as an asymptotic stationary proxy for admissibility curvature, giving it the status of a sufficient statistic for the admissibility geometry of the fiber, rather than a fundamental dynamical quantity. In Section 7, we derived the exponential weight $e^{iS/\hbar}$ from the oscillatory structure of the complexified admissibility functional, and derived $\hbar$ as the emergent phase-coherence scale (29.12). In Section 8 (Theorem 29.1), we proved that classical paths dominate because they correspond to stationary points of admissibility curvature, deriving the Euler-

Lagrange equations as the stationarity condition for admissibility curvature and classical mechanics as the leading-order term in a $1/\kappa_A$ expansion. In Section 22 (Theorem 29.1 formal version), we provided the complete axiomatic reconstruction of the standard path integral from Phase Theory axioms through a six-step procedure. In Section 25 (Theorem 29.2), we established the Empirical Equivalence Theorem: Phase Theory reproduces all predictions of standard QFT in the smooth-admissibility regime.

Five major conceptual gains follow from this framework. First, the many-worlds reading of the path integral is dissolved (Section 9): paths are projection labels, not histories, and the ontological proliferation of Everett branches is an artifact of confusing the mathematical integration variable with a physical entity. Second, the primacy of the action is explained (Section 5): $S[x]$ is privileged not because it is fundamental but because it is the most efficient scalar summary of the admissibility geometry of the fiber $\mathcal{P}(x)$ that is computable from the coarse-grained description. Third, the measurement problem is resolved (Section 20): measurement is admissibility pruning, "collapse" is the restriction of $C\Phi^{\text{adm}}$ by an additional constraint ACF_{meas} , and the Born rule is derived from the admissibility measure ratio (29.28). Fourth, renormalization is clarified (Section 14): UV divergences are overcounting of inadmissible sub-coherence-scale configurations, and the RG flow is the flow of the admissibility coarse-graining projection as the resolution scale changes. Fifth, quantum gravity path integrals are rendered conceptually well-defined and UV-finite (Section 19): geometries are coarse-grained phase projections, the sum over geometries is a weighted count of admissible configurations projecting to different metric descriptions, and GCE prunes all sub-Planck fluctuations, removing the UV divergences of perturbative quantum gravity.

Eight classes of falsifiable predictions beyond standard QFT have been enumerated in Section 26, ranging from precision tests of the particle

spectrum to gravitational wave propagation anomalies, dark-sector recoil signatures, and quantum gravity amplitude UV-finiteness. These predictions distinguish Phase Theory from all known alternatives in experimentally accessible regimes, and their confirmation or falsification will determine whether Phase Theory has correctly identified the ontological structure underlying quantum mechanics and quantum field theory.

We close with the philosophical moral that unifies the entire analysis. The path integral works because phase structure is rich — richer than any coarse-grained description can capture — and our descriptions must compress it. The sum over paths is the most efficient compression of the admissibility geometry of $C\Phi$ that is available at the level of a coarse-grained reduced description. Its power comes not from any mystical summation over existing histories, not from any branching of worlds, and not from any fundamental role of the Lagrangian or the action. It comes from the fact that nature's phase configuration space has a specific, highly organized admissibility structure — enforced globally by GCE — and the path integral is the natural integral transform that makes that structure computationally accessible. *Nature does not integrate paths. We do — because phase structure is rich, and our descriptions must compress it.*

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Appendix A: Notation Summary

Symbol	Definition / Meaning	First Appears
M	Topological substrate (compact, connected)	Section 2
Φ	Phase field — section of $P(M, U(1))$	Section 2, Def. 29.1
$C\Phi$	Space of all phase configurations (Fréchet manifold)	Section 2, Def. 29.1
$C\Phi^{\text{adm}}$	Admissible configuration space ($A[\Phi] = 1$)	Section 2, Def. 29.2
$A[\Phi]$	Admissibility functional: $C\Phi \rightarrow [0,1]$	Section 2, Def. 29.2
GCE	Global Consistency Enforcement	Section 2, Def. 29.2
$\kappa_A[\Phi]$	Admissibility curvature: $-\Delta_c \log A[\Phi]$	Section 2, Eq. (29.1)
Π_λ	Coarse-graining projection map $C\Phi \rightarrow D_\lambda$	Section 2, Eq. (29.2)
D_λ	Reduced description space at resolution λ	Section 2
$\langle O \rangle$	Admissibility-weighted average of observable O	Section 2, Eq. (29.3)
$\mathcal{P}(x)$	Phase-admissible path class for projection label x	Section 3, Def. 29.3
ACF	Admissibility Constraint Functional	Section 4, Def. 29.4
$C\Phi^{\text{adm}}(x_i, x_f)$	Boundary-constrained admissible configuration space	Section 4, Def. 29.4
μ_{path}	Derived path measure	Section 4, Eq. (29.5)

Symbol	Definition / Meaning	First Appears
	on D	
$S[x]$	Effective action (admissibility proxy)	Section 5, Def. 29.5
$\Psi_A[\Phi]$	Admissibility phase of configuration Φ	Section 5, Def. 29.5
$K(x_f, t_f; x_i, t_i)$	Admissibility Overlap Kernel	Section 6, Def. 29.6
$A_C[\Phi]$	Complexified admissibility functional	Section 7, Eq. (29.9)
$\Theta_A[\Phi]$	Admissibility phase (real-valued)	Section 7, Eq. (29.9)
$\hbar$	Phase coherence scale (derived, not postulated)	Section 7, Eq. (29.12)
ρ_{adm}	Density of admissible configurations per unit phase-bundle volume	Section 7, Eq. (29.12)
V_{coh}	Phase coherence volume of the substrate	Section 7, Eq. (29.12)
$Z[J]$	Generating functional of QFT	Section 10, Eq. (29.15)
$W[J]$	Connected generating functional $W = -i\hbar \log Z$	Section 10, Eq. (29.16)
$Q[\Phi]$	Topological charge (winding number) of configuration	Section 13
AB	Admissibility Basin (Definition 29.7)	Section 13, Def. 29.7
S_{inst}	Instanton action (admissibility barrier measure)	Section 13, Eq. (29.19)
S_λ	Wilsonian effective action at scale λ	Section 14
ℓ_{coh}	Phase coherence length of the substrate	Section 14
ACF_{meas}	Measurement	Section 20, Eq. (29.27)

Symbol	Definition / Meaning	First Appears
	admissibility constraint functional	
$P(o)$	Born-rule probability of measurement outcome o	Section 20, Eq. (29.28)
$g_{\mu\nu}$	Emergent metric (coarse-grained projection of Φ)	Section 19, Eq. (29.26)
λ_{Pl}	Planck resolution scale (admissibility limit)	Section 19
T_{sat}	Phase saturation temperature	Section 26
η/s	Viscosity-to-entropy ratio (of QGP)	Section 26

Appendix B: Admissibility Functional — Formal Axiomatic Definition

We provide the complete axiomatic definition of the admissibility functional $A : C\Phi \rightarrow [0,1]$ used throughout this paper.

Axiom B1 (Domain). The domain of A is the Fréchet manifold $C\Phi = \Gamma(P(M, U(1)))$ of smooth sections of the principal phase bundle over the compact topological substrate M . The Fréchet manifold structure is equipped with the $L^2(M)$ metric topology on sections.

Axiom B2 (Range). $A : C\Phi \rightarrow [0,1]$. The value $A[\Phi] = 1$ represents full admissibility (perfect global consistency); $A[\Phi] = 0$ represents complete inadmissibility (maximal global inconsistency). The range $[0,1]$ is exact: no extension beyond this interval is permitted.

Axiom B3 (GCE Condition). $A[\Phi] = 1$ if and only if Φ is a globally consistent phase configuration satisfying all GCE constraints. That is, $C\Phi^{adm} := A^{-1}(\{1\})$ is the set of all fully admissible configurations, and this set is non-empty

(guaranteed by the existence of at least one consistent global phase configuration — a structural axiom of Phase Theory).

Axiom B4 (Log-Concavity). The functional A is log-concave on $C\Phi$: for any $\Phi_1, \Phi_2 \in C\Phi$ and any $t \in [0,1]$,

$$\log A[(1-t)\Phi_1 + t\Phi_2] \geq (1-t) \log A[\Phi_1] + t \log A[\Phi_2] \quad (\text{B.1})$$

Log-concavity ensures that admissibility basins are convex in the logarithmic sense, which is necessary for the fiber integrals of Section 7 to converge and for the Gaussian approximation to be well-controlled.

Axiom B5 (Locality in the Coarse-Grained Regime). In the smooth-admissibility regime ($A[\Phi]$ near unity, $\kappa_A[\Phi]$ finite), A admits a local decomposition:

$$A[\Phi] = \exp \left(- \int_M \lambda_A(m, \Phi(m), \nabla \Phi(m), \nabla^2 \Phi(m)) dm \right) \quad (\text{B.2})$$

where $\lambda_A(m, \Phi, \nabla \Phi, \nabla^2 \Phi)$ is the local admissibility density, a smooth function of its arguments. This locality axiom is the Phase Theory analog of the locality of the Lagrangian in standard QFT, and it is the structural condition that allows the derivation of the action functional from the admissibility functional in the smooth regime (Proposition 29.3).

Axiom B6 (Topological Invariance). $A[\Phi]$ is constant on connected components of $C\Phi^{\text{adm}}$ with fixed topological charge $Q[\Phi]$. That is, A depends on the homotopy class of Φ but not on the specific representative within a homotopy class of fully admissible configurations. This axiom encodes the fact that GCE enforces global topological consistency, not just local

consistency, and is the foundation of the topological effects discussed in Section 18.

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